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MITIGACIÓN DEL OLVIDO CATASTRÓFICO EN CLASIFICACIÓN DE OBJETOS MEDIANTE REPRESENTACIONES GAUSSIANAS DE CLASE

El presente trabajo aborda el problema del aprendizaje continuo (*Continual Learning*) en clasificación de objetos, con el objetivo de mitigar el olvido catastrófico (*catastrophic forgetting*) que sufren los modelos profundos tradicionales cuando se les expone a flujos de datos no estacionarios. Se propone un esquema modular que desacopla la extracción de características de la clasificación: un extractor de características pre-entrenado y congelado (DINOv2) provee una representación general del espacio visual, mientras que cada clase de objeto se modela mediante una distribución Gaussiana independiente en dicho espacio. La clasificación se realiza por máxima verosimilitud (regla MAP), lo que permite incorporar nuevas clases sin necesidad de re-entrenar la red ni de almacenar muestras antiguas.

Se estudian sistemáticamente distintos estimadores de la matriz de covarianza (diagonal, completo, híbrido, compartido, Ledoit–Wolf y OAS), variantes de Mixturas de Gaussianas (MoG) con 2 y 4 componentes, y una familia de diez estrategias de fusión generativa–discriminativa y de re-ordenamiento que combinan la verosimilitud Mahalanobis con la similaridad angular del coseno, fusión aditiva en log-dominio, normalización min–max y escalamiento por temperatura. Adicionalmente, se evalúa la calibración del modelo mediante el Expected Calibration Error (ECE), Total Confidence, Confidence on Errors, Log-Gap y entropía predictiva, y se valida la hipótesis Gaussiana a través de un análisis univariado de asimetría y curtosis sobre los *embeddings* y de la comparación de las distancias de Mahalanobis empíricas con la distribución teórica χ^2.

Los experimentos se realizan sobre cuatro *benchmarks* de aprendizaje incremental por clases: Split CIFAR-100 (10 tareas × 10 clases), Split ImageNet-R (10 tareas × 20 clases), Split Caltech-256 (10 tareas × 25 clases, utilizando sólo las primeras 250 clases) y un *benchmark* unificado de 30 tareas (Split Joint Datasets) que concatena los tres anteriores en un único flujo heterogéneo. Los resultados muestran que el método propuesto alcanza desempeños competitivos con el estado del arte basado en *prompts* (*e.g.* L2P, DualPrompt, CODA-Prompt, RainbowPrompt y APT), llegando incluso a superarlos en escenarios de alta heterogeneidad de tareas como Caltech-256 y Joint, manteniendo a la vez un olvido reducido (por debajo del 5 % en los *benchmarks* de 10 tareas) y una arquitectura significativamente más liviana, escalable y libre de gradientes que sus competidores. El análisis comparativo identifica al estimador Híbrido de covarianza y a las estrategias aditivas de fusión en log-dominio (SoftTaskFus, OptFusion-MM y TempScale) como las elecciones más robustas y recomendables para el despliegue práctico del marco.

MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

This work addresses the problem of Continual Learning for object classification, with the goal of mitigating the catastrophic forgetting that affects traditional deep models when they are exposed to non-stationary data streams. A modular framework is proposed in which feature extraction is decoupled from classification: a frozen, pre-trained DINOv2 feature extractor provides a general representation of the visual space, while every object class is modelled by an independent Gaussian distribution in that space. Classification is performed via maximum a posteriori (MAP) inference, which allows new classes to be incorporated without re-training the network or storing past samples.

The thesis investigates several covariance estimators (diagonal, full, hybrid, shared, Ledoit–Wolf and OAS), two- and four-component Mixtures of Gaussians (MoG) and a family of ten generative–discriminative fusion and re-ranking strategies that combine the Mahalanobis log-likelihood with cosine angular similarity, additive log-domain fusion, min–max normalisation and temperature scaling. Probabilistic calibration is evaluated through the Expected Calibration Error (ECE), Total Confidence, Confidence on Errors, Log-Gap and predictive entropy, and the Gaussian assumption is validated through a univariate analysis of skewness and kurtosis of the embeddings as well as through the comparison of empirical squared Mahalanobis distances against the theoretical χ^2 density.

Experiments are conducted on four class-incremental benchmarks: Split CIFAR-100 (10 tasks $\times$ 10 classes), Split ImageNet-R (10 tasks $\times$ 20 classes), Split Caltech-256 (10 tasks $\times$ 25 classes, using only the first 250 classes) and a 30-task unified benchmark (Split Joint Datasets) that concatenates the three previous benchmarks into a heterogeneous stream. The results show that the proposed method achieves performance competitive with the prompt-based state of the art (*e.g.* L2P, DualPrompt, CODA-Prompt, RainbowPrompt and APT), and overtakes it on the most heterogeneous benchmarks (Caltech-256 and Joint), while keeping forgetting below 5 % on the 10-task benchmarks and offering a substantially lighter, gradient-free and more scalable architecture than its prompt-based competitors. The comparative analysis identifies the Hybrid covariance estimator and the additive log-domain re-ranking strategies (SoftTaskFus, OptFusion-MM and TempScale) as the most robust and recommended choices for practical deployment of the framework.

Abstract

Las arquitecturas tradicionales de aprendizaje profundo son predominantemente monolíticas y suelen fallar al ser expuestas a distribuciones de datos no estacionarias. Incorporar nuevas clases requiere por lo general un re-entrenamiento exhaustivo sobre el conjunto completo de datos para evitar el olvido catastrófico, lo que limita su despliegue en sistemas dinámicos del mundo real. Para enfrentar estos desafíos, este trabajo presenta un esquema modular de Aprendizaje Continuo (*Continual Learning*, CL) que desacopla la extracción de características de la clasificación. Utilizando un extractor de características pre-entrenado y congelado (DINOv2) en conjunto con modelos Gaussianos independientes por clase, se construyen representaciones probabilísticas distintas para cada categoría de objeto. Este desacoplamiento arquitectónico permite la incorporación incremental de nuevas categorías sin interferir con el conocimiento previamente adquirido. Una evaluación extensa sobre cuatro *benchmarks* —Split CIFAR-100, Split ImageNet-R, Split Caltech-256 (250 clases, 10 tareas) y el Split Joint Datasets de 30 tareas— demuestra que el método alcanza desempeño competitivo con los enfoques del estado del arte basados en *prompts*, e incluso los supera en los escenarios más heterogéneos. Adicionalmente, los resultados destacan que la estrategia propuesta es significativamente más escalable, computacionalmente liviana y simple de implementar que los métodos basados en *replay* o regularización intensiva, ofreciendo así una solución robusta para el aprendizaje continuo eficiente.

Abstract

Traditional deep learning architectures are predominantly monolithic and often fail to adapt when exposed to non-stationary data distributions. Incorporating new classes typically requires exhaustive re-training on the full dataset to prevent catastrophic forgetting, a constraint that limits their deployment in dynamic, real-world systems. To overcome these challenges, this work introduces a modular Continual Learning (CL) framework that decouples feature extraction from classification. By pairing a frozen, pre-trained DINOv2 feature extractor with independent class-conditional Gaussian models, the framework builds distinct probabilistic representations for every object category. This architectural decoupling enables the seamless, incremental addition of new categories without interfering with previously acquired knowledge. An extensive evaluation on four benchmarks —Split CIFAR-100, Split ImageNet-R, Split Caltech-256 (250 classes, 10 tasks) and the 30-task Split Joint Datasets— demonstrates that the method achieves performance competitive with prompt-based state-of-the-art approaches, and overtakes them on the most heterogeneous scenarios. The proposed strategy is also significantly more scalable, computationally lighter and simpler to implement than replay- or regularisation-intensive methods, offering a robust solution for efficient continual learning.

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Chapter 1

Introduction

1.1 Motivation

State-of-the-art (SOTA) object classification methods rely on deep learning models that are either trained end-to-end on the target application or initially trained on one task and subsequently fine-tuned for a different one. In both cases the resulting models are *monolithic*: a single network is trained on a large dataset and performs feature extraction and classification jointly. Such models are highly effective when the set of object classes is fixed and known at training or fine-tuning time. When new object classes must be incorporated, however, the entire model must be retrained or fine-tuned. This procedure requires significant computational resources, demands access to the original training dataset and, in many cases, leads to *catastrophic forgetting* of previously learned classes [6].

A natural way to integrate new classes while avoiding catastrophic forgetting is to formulate the problem as continual or class-incremental classification, in which the algorithm must learn to distinguish an ever-growing number of categories [7]. In this paradigm, a *task* corresponds to a distinct portion of the training experience, presented sequentially to the model. Instead of having access to the entire dataset at once, as in conventional supervised learning, the model receives data in sequential chunks [8]. Each task typically corresponds to a set of new classes that must be recognised. The central challenge is that when the model is updated task by task, it tends to overwrite previously acquired knowledge: monolithic models are structurally ill-suited for this scenario, and an alternative paradigm is therefore required.

This limitation has motivated the SOTA in continual classification to focus on balancing *stability* and *plasticity*. Stability refers to a model's ability to preserve previously learned knowledge; plasticity describes its capacity to adapt and incorporate new tasks [9]. Traditional approaches rarely optimise both properties simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones and by the need to continually adjust the network's structure to match the growing number of object classes.

A natural way to overcome this dilemma is to decouple the monolithic model into two distinct components that are trained separately, possibly at different times: one responsible for feature extraction and another for object classification. The feature-extraction model must be trained at system-initialisation time and must produce embeddings that are general enough to characterise every class of interest, including classes unknown at training time. The classification model, by contrast, should be easy to update or extend as new object classes arrive, and ideally should be able to incorporate a new class without modifying the components that already recognise the existing classes. This structural separation mirrors the organisation of the human visual system, in which the structures responsible for low-level feature extraction (simple and complex cells in cortical areas V1 and V2) develop during the early post-natal weeks and remain largely stable thereafter [10]. High-level object recognition, in contrast, occurs in cognitive areas where the ability to learn new categories persists throughout life, occasionally producing slight degradation in the recognition of visually similar classes.

The emergence of large-scale, self-supervised visual foundation models (such as DINOv2 [23]) provides the empirical conditions to put this idea into practice. Foundation models offer general-purpose feature spaces with high transferability, and once frozen they yield a stationary representation in which incremental learning can be reduced to estimating class-conditional statistics rather than adapting network parameters. Building on this observation, we propose a continual learning object recognition system organised as two complementary components. The first component is a feature extractor built from a frozen pre-trained foundation model. The second component is a statistical classifier built from a collection of class-conditional Gaussian models, where each model characterises a specific object class independently of the others. A new Gaussian model can be created and inserted whenever a new class arrives, and classification is performed by assigning the input to the class whose Gaussian yields the highest log-likelihood, in direct analogy with a Bayes classifier.

Gaussian class representations offer two key advantages. First, each class is modelled independently from the rest, so incorporating a new class amounts to estimating a small number of sufficient statistics (mean and covariance) and does not require any modification of the previously stored class models. Second, Gaussian models are compact: they encode each class in a few parameters that can be estimated from a modest sample budget and refined as new samples become available, without any storage of the underlying images.

1.2 Hypothesis

This thesis hypothesises that the combination of an independent class-conditional Gaussian model and a frozen pre-trained feature extractor provides a robust framework for continual object classification. By enabling the addition of new classes without full-model retraining, this design eliminates the conventional source of catastrophic forgetting, while retaining a simple, compact and computationally inexpensive architecture whose accuracy and forgetting are competitive with the state of the art.

1.3 General Objective

The main objective of this thesis is to develop and validate a continual learning framework for object classification that uses pre-trained foundation models as feature extractors and independent class-conditional Gaussian models as classifiers, with the purpose of enabling the incremental incorporation of new classes while preventing catastrophic forgetting of previously learned ones.

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- To assess the Gaussianity of the class-conditional feature distributions in the DINOv2 latent space, validating the underlying distributional assumptions of the proposed model.
- To characterise the efficacy of independent Gaussian distributions as class representations within a continual-learning framework, and to compare the result against prompt-based state-of-the-art baselines.
- To quantify the impact of various covariance-matrix estimators (diagonal, full, hybrid, shared, Ledoit–Wolf, OAS) and Mixture-of-Gaussian variants on the classification accuracy, the average forgetting and the probabilistic calibration of the resulting classifier.
- To design and evaluate a family of generative–discriminative fusion and re-ranking strategies that combine the Mahalanobis log-likelihood with auxiliary discriminative cues, and to identify which of them improve the baseline reliably and which are brittle.
- To analyse the probabilistic calibration of the proposed model through a comprehensive set of metrics (Expected Calibration Error, Confidence on Errors, Total Confidence, Log-Gap, predictive entropy) that go beyond raw accuracy.
- To evaluate the dependence of the framework on the choice of upstream feature extractor by comparing DINOv2 against supervised Vision Transformers of varying capacity (ViT-base/16, ViT-medium/16) across multiple benchmarks and incremental-learning scenarios.

1.5 Contributions

The principal contributions of this thesis are the following:

1. A modular, gradient-free Bayesian continual-learning framework that decouples feature extraction from classification by combining a frozen DINOv2 backbone with in-

dependent class-conditional Gaussian models, achieving competitive accuracy and low forgetting without any rehearsal mechanism.

2. A systematic comparison of covariance estimators in the high-dimensional, small-sample regime, including the introduction of a *Hybrid* estimator that decouples the volume term (computed on the diagonal) from the shape term (computed on the full inverse). The Hybrid estimator is shown to dominate every alternative on the heavy-tailed and long-horizon benchmarks.
3. A family of ten re-ranking and fusion strategies that integrate generative log-likelihood evidence with discriminative cosine similarity, log-domain task priors and temperature-scaled posteriors. An empirical taxonomy is derived from the results, separating these strategies into *recommended*, *brittle* and *marginal*.
4. A complete probabilistic-calibration analysis of the framework, including ECE, Total Confidence, Confidence on Errors, Log-Gap and predictive entropy across covariance estimators and benchmarks, with explicit identification of the volume–shape trade-off as the structural cause of the over-confidence observed under the Full-Covariance branch.
5. A backbone-sensitivity study (DINOv2, ViT-base, ViT-medium) and a 30-task heterogeneous Joint benchmark that aggregates CIFAR-100, ImageNet-R and the first 250 categories of Caltech-256 into a single stream, providing the most demanding evaluation regime currently reported for closed-form Bayesian continual classifiers.

1.6 Thesis Structure

The remainder of the document is organised as follows.

Chapter 2 – Theoretical Framework. Establishes the mathematical and conceptual basis of the work. It reviews the relevant subset of artificial intelligence, deep learning and computer vision, presents Bayesian classification and the Mixture-of-Gaussian extension, introduces the high-dimensional covariance estimation tools (regularisation, Ledoit–Wolf and OAS shrinkage), discusses the empirical Gaussianity diagnostics (skewness, kurtosis, Mahalanobis distance) and surveys the continual-learning paradigm together with its dominant mitigation strategies.

Chapter 3 – Methodology. Describes the proposed architecture and its components in detail. It defines the experimental protocol for the four benchmarks —Split CIFAR-100, Split ImageNet-R, Split Caltech-256 (10 tasks of 25 classes each over the first 250 categories) and the 30-task Joint stream—, formalises the six covariance estimators and the Mixture-of-Gaussian variants, derives the ten generative–discriminative fusion and re-ranking strategies, discusses temperature-scaled Bayesian calibration and introduces the probabilistic metrics used in the evaluation.

Chapter 4 – Results. Reports the quantitative results of every experiment carried out in the thesis. It is structured by the same top-down logic used to validate the modelling pipeline: per-dimension Gaussianity moments, feature-space diagnostics (PCA spectral decay and Mahalanobis histograms), continual-learning metrics across covariance estimators, generative–discriminative fusion tables, backbone ablations, calibration metrics, and incremental class-inclusion curves. All interpretation is deferred to the next chapter.

Chapter 5 – Discussion and Analysis. Interprets the experimental evidence. It examines the statistical characterisation of the DINOv2 embedding space, compares the covariance estimators and mixtures under the volume–shape decomposition principle, evaluates the reranking family and identifies which strategies improve the baseline reliably, and discusses backbone sensitivity, calibration and the framework's computational footprint. The chapter closes with a synthesis of the principal findings and a practical recipe.

Chapter 6 – Conclusion. Summarises the main conclusions of the work and outlines two natural extensions of the framework: open-world discovery (using the model's confidence as an out-of-distribution score) and real-time object detection (leveraging the spatial tokens of the DINOv2 backbone to avoid an external region-proposal network).

Chapter 2

Theoretical Framework

This chapter establishes the conceptual and mathematical foundations required to contextualize this research, transitioning from broad paradigms in Artificial Intelligence to the specific statistical mechanics and adaptive learning strategies underpinning our methodology. We begin by reviewing the fundamental principles of Machine Learning, specifically within the realms of image processing and object classification, before providing a rigorous mathematical treatment of Bayesian Inference, Mixture of Gaussians (MoG) and Gaussian Analysis. Furthermore, we detail advanced covariance estimation methods such as OAS and Ledoit-Wolf, which are critical for stability in high-dimensional feature spaces. Finally, we address the challenge of "catastrophic forgetting" by introducing the Continual Learning (CL) paradigm, describing the datasets utilized and evaluating the state-of-the-art mitigation strategies that provide a performance baseline for the proposed work.

2.1 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a broad field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and language understanding [11]. Within this domain, Machine Learning (ML) focuses on developing algorithms that allow computers to learn patterns from data without explicit programming. ML systems are generally categorized into three paradigms: Supervised Learning, Unsupervised Learning, and Reinforcement Learning [12].

First, in **Supervised Learning**, the model learns a mapping from input variables $\mathbf{x}$ to output variables y based on a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$. The objective is to minimize a loss function that penalizes the discrepancy between the predicted output and the ground truth. This paradigm is typically divided into Classification, where the output is a discrete category, and Regression, where the output is a continuous value.

Second, **Unsupervised Learning** deals with unlabeled data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$. Instead of predicting a target, the goal is to discover the underlying structure or distribution of the data. Common tasks include Clustering, which groups similar data points (e.g., K-Means or

Gaussian Mixture Models), and Dimensionality Reduction, which projects high-dimensional data into a lower-dimensional space while preserving essential information (e.g., PCA or t-SNE).

Finally, **Reinforcement Learning** is based on an agent interacting with an environment to maximize a cumulative reward. Unlike supervised learning, the agent is not told which actions to take but must discover them through trial and error. This process is often modeled as a Markov Decision Process (MDP) [13], where the agent transitions between states based on actions, receiving feedback in the form of numerical rewards. The focus is on finding an optimal policy that balances exploration (trying new actions) and exploitation (using known high-reward actions).

2.1.1 Deep Learning

In traditional Machine Learning, the feature extraction step is a manual process where a human expert must identify and define the most relevant characteristics (such as the shape or edges of the car) before passing them to a classification algorithm. In contrast, as shown in figure 2.1, Deep Learning utilizes neural networks to automate this process; the model performs both feature extraction and classification simultaneously, learning to identify complex patterns directly from the input data without human intervention.

Structurally, this shift allows Deep Learning to handle much more complex, high-dimensional data, though it typically requires significantly larger datasets and more computational power to be effective. While traditional Machine Learning depends on the quality of hand-crafted features, Deep Learning architectures—like multi-layered neural networks—are designed to build hierarchical representations of data, progressively moving from low-level pixels to high-level object concepts.

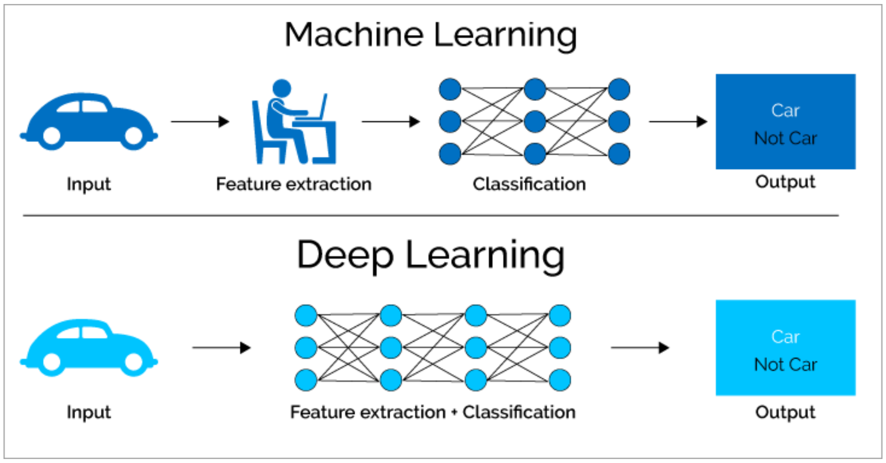


Figure 2.1: Machine Learning and Deep Learning Strategies. Image taken from [1].

2.1.2 Computer Vision and Vision Transformers

Computer Vision is the interdisciplinary subfield of AI that enables computers to gain high-level understanding from digital images or videos, effectively automating tasks performed by the human visual system [14]. While Convolutional Neural Networks (CNNs) [15] historically dominated this field, the Transformer architecture—originally developed for Natural Language Processing—has revolutionized visual recognition through the self-attention mechanism [16].

Vision Transformers (ViT) adapt this architecture by treating an image as a sequence of discrete tokens. As illustrated in Figure 2.2, the process follows a specific pipeline:

- **Patch Partitioning:** The input image is decomposed into fixed-size patches.
- **Linear Projection:** These patches are flattened and mapped to an embedding space.
- **Positional Encoding:** Since Transformers lack inherent knowledge of spatial structure, learnable position embeddings are added to the patch embeddings.
- **Transformer Encoder:** The resulting sequence is processed through multiple layers of multi-head self-attention and MLP blocks to capture global dependencies.

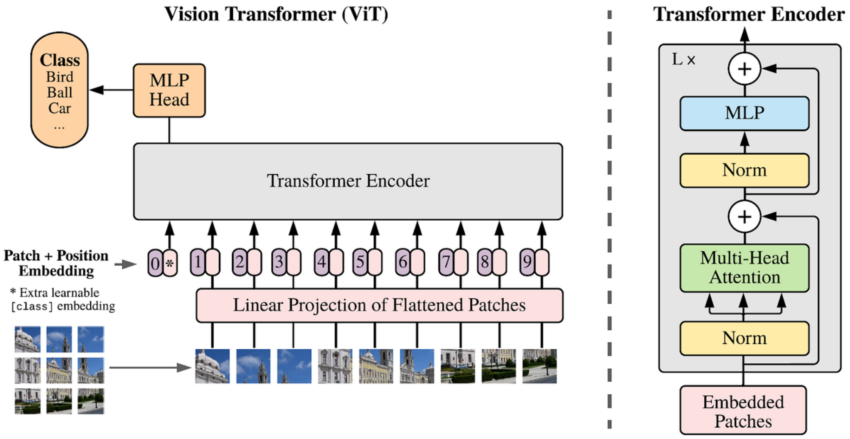

Vision Transformer (ViT)
Class
Bird
Ball
Car
...
MLP Head
Transformer Encoder
Patch + Position Embedding
* Extra learnable [class] embedding
0 *
1
2
3
4
5
6
7
8
9
Linear Projection of Flattened Patches
Transformer Encoder
L x
+
MLP
Norm
+
Multi-Head Attention
Norm
Embedded Patches

Figure 2.2: Vision Transformer (ViT) architecture overview. The model processes an image as a sequence of patches through a standard Transformer encoder. Image taken from [2].

Object Classification and Representative Power

The primary objective of the original ViT was *Object Classification*: the fundamental task of assigning a single categorical label to an entire input image. This task serves as the primary benchmark for evaluating the representative power of visual models [17]. By processing the global sequence of patches, ViTs have demonstrated an ability to meet or exceed CNN performance, particularly when pre-trained on the large-scale datasets described below.

2.1.3 Image Datasets and Benchmarks

To rigorously test the robustness and generalization of architectures like ViT, this work utilizes several standardized datasets that vary in scale, stylistic diversity, and granularity.

The ImageNet Family

ImageNet [18] is a cornerstone of computer vision, providing over 1.2 million images across 1,000 classes. Its hierarchical structure, based on WordNet, makes it the standard for large-scale classification. To further push the boundaries of model performance, two variants are often employed:

- **ImageNet-21K (and ImageNet-21k-P):** An expansive version containing 14 million images across 21,000 classes [19]. The "P" version is a cleaned subset designed to make pre-training for models like ViT more accessible.
- **ImageNet-R (Renditions):** A dataset of 30,000 images covering 200 classes in non-photographic styles (e.g., art, sketches). It tests a model's robustness to out-of-distribution data [20].

Fine-Grained and Diverse Benchmarks

Beyond large-scale pre-training, fine-grained classification is evaluated using:

- **CIFAR-100:** Consists of 60,000 low-resolution (32×32) images [21]. It is unique for its two-level hierarchy, where 100 "fine" classes are nested within 20 "coarse" superclasses, allowing for the evaluation of both specific and broad categorization.
- **Caltech-256:** Provides 30,607 images across 256 categories [22]. It offers greater category diversity and a higher minimum number of images per class than its predecessors, serving as a robust benchmark for generalisation. In this thesis we use the first 250 categories (approximately 28,100 images) so that the resulting class-incremental protocol can be divided into 10 balanced tasks of 25 classes each (see Chapter 3).

Self-Supervised Advancements: DINOv2

While the aforementioned datasets provide explicit labels for supervision, recent advancements have led to foundation models such as **DINOv2** [23]. Unlike earlier iterations, DINOv2 utilizes a discriminative self-supervised approach that combines contrastive learning with image-level and patch-level objectives. By training on massive, curated datasets without human labels, DINOv2 produces multipurpose visual features that exhibit strong "out-of-the-box" properties, often surpassing models trained with explicit supervision on tasks like semantic segmentation [23].

2.2 Foundations of Probabilistic Modeling

The convergence of statistical learning theory and high-dimensional data analysis has necessitated the development of robust mathematical frameworks to handle the dual challenges of uncertainty quantification and dimensionality reduction. As modern neural architectures, such as the DINOv2 Vision Transformer, extract increasingly complex representations from datasets like ImageNet-R, the subsequent classification and density estimation layers must be grounded in rigorous probabilistic foundations.

2.2.1 Bayes Classification

At its core, Bayes classification is rooted in **Bayes' Theorem**, which provides a mathematical framework for updating the probability of a hypothesis based on observed evidence. For a classification task, we seek the posterior probability $p(y = c \mid \mathbf{x})$—the probability that an observation belongs to class c given its observed feature vector $\mathbf{x}$.

Bayes' Theorem is defined as:

$$p(y = c \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid y = c)\, p(y = c)}{p(\mathbf{x})} \tag{2.1}$$

where:

- $p(y = c \mid \mathbf{x})$ is the **posterior** probability.
- $p(\mathbf{x} \mid y = c)$ is the **likelihood** of the data given the class.
- $p(y = c)$ is the **prior** probability of the class.
- $p(\mathbf{x})$ is the **evidence** or marginal likelihood, acting as a normalization constant.

The objective of a Bayes classifier is to minimize the Bayes risk; in the simplest case, this corresponds to assigning the observation to the class that maximizes the posterior probability.

2.2.2 Likelihood-Based Classification

Likelihood inference provides a coherent framework for updating the probability of a hypothesis as more evidence or information becomes available. In contrast to frequentist approaches that treat parameters as fixed but unknown constants, the Bayesian paradigm treats parameters as random variables characterized by probability distributions.

The connection between Bayes' Theorem and likelihood-based classification lies in the **Maximum A Posteriori (MAP)** decision rule. Since the evidence $p(\mathbf{x})$ is constant across all classes for a given observation, the classification decision depends solely on the product of the likelihood and the prior:

$$p(y = c \mid \mathbf{x}) \propto p(\mathbf{x} \mid y = c)\, p(y = c). \tag{2.2}$$

A generative modeling framework can be adopted and class-conditional Gaussian feature distributions can be assumed:

$$p(\mathbf{x} \mid y = c) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{2.3}$$

where $\mathbf{x} \in \mathbb{R}^d$ represents the feature vector, $\boldsymbol{\mu}_c$ is the class mean vector, and $\boldsymbol{\Sigma}_c \succ 0$ is the covariance matrix.

Classification can be performed via MAP inference by maximizing the log-posterior:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\mathbf{x} \mid y = c) + \log \pi_c\right], \tag{2.4}$$

where $\pi_c = p(y = c)$ denotes the class prior.

Substituting the multivariate Gaussian density into the MAP rule, the decision rule becomes equivalent to:

$$\hat{y} = \arg\min_{c} \left[(\mathbf{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| - 2 \log \pi_c\right]. \tag{2.5}$$

Mixture of Gaussians (MoG)

A Mixture of Gaussians models a complex probability density as a convex combination of K distinct Gaussian components. This approach is fundamental for clustering and density estimation in multi-modal distributions. Therefore, to capture multimodal feature distributions, the class-conditional model can be extended to a mixture:

$$p(\mathbf{x} \mid y = c) = \sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}), \tag{2.6}$$

where $\alpha_{c,k} \geq 0$ and $\sum_k \alpha_{c,k} = 1$.

The MAP decision rule becomes

$$\hat{y} = \arg\max_{c} \left[\log \left(\sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k})\right) + \log \pi_c\right]. \tag{2.7}$$

2.2.3 Advanced Covariance Estimation in High Dimensions

In many modern applications, such as image feature analysis using DINOv2, the dimensionality d (often in the hundreds or thousands) often exceeds the sample size n. In this “large d, small n” regime, the empirical sample covariance matrix (SCM) S is a poor estimator: its eigenvalues are biased, and the matrix is singular if $n < d$. This singularity prevents the inversion of the matrix required for the quadratic form in classification and density estimation.

Regularization and Numerical Stability

To stabilize the inversion of the covariance matrix $\mathbf{\Sigma}_c$ for each class, regularization techniques are employed. A primary method is diagonal loading, which adds a small constant ε (e.g., 10^{-6}) to the diagonal elements:

$$\mathbf{\Sigma}_{reg} = \mathbf{\Sigma}_c + \varepsilon I \tag{2.8}$$

This adjustment ensures that the matrix remains strictly positive definite. However, for higher numerical precision in calculating the inverse (precision matrix) and the log-determinant, we utilize Spectral Filtering via eigendecomposition.

Spectral Filtering and Precision Matrix Inversion

For each class, the system computes the eigenvectors P and eigenvalues λ such that $\mathbf{\Sigma}_{reg} = P\Lambda P^T$. To ensure stability against near-zero eigenvalues that survive initial regularization, we apply a floor: $\hat{\lambda} = \max(\lambda, \varepsilon)$. The precision matrix (inverse covariance) is then reconstructed as:

$$\mathbf{\Sigma}_c^{-1} = P\text{diag}\left(\frac{1}{\hat{\lambda}_1}, \ldots, \frac{1}{\hat{\lambda}_d}\right) P^T \tag{2.9}$$

The log-determinant, required for the Gaussian log-likelihood, is computed directly from these filtered eigenvalues to avoid underflow:

$$\ln|\mathbf{\Sigma}_c| = \sum_{j=1}^{d} \ln(\hat{\lambda}_j) \tag{2.10}$$

Finally, the normalization constant $\ln A_c$ (representing the constant term of the multivariate Gaussian log-density) is stored:

$$\ln A_c = -\frac{1}{2}\left(d\ln(2\pi) + \ln|\mathbf{\Sigma}_c|\right) \tag{2.11}$$

This approach eliminates “noise floor” issues that would otherwise cause the matrix to become singular or produce infinite values during log-likelihood estimation.

Linear Shrinkage Estimators

Linear shrinkage methods propose a compromise between the SCM S and a highly structured target matrix F, such as the identity matrix or a diagonal matrix of sample variances.

Ledoit-Wolf Estimator The Ledoit-Wolf estimator [24] determines an optimal shrinkage intensity ρ to minimize the expected Frobenius norm between the estimate and the true covariance matrix $\mathbf{\Sigma}$. It is defined as:

$$\hat{\mathbf{\Sigma}}_{LW} = (1 - \rho)S + \rho F \tag{2.12}$$

Ledoit-Wolf is computationally efficient and provides an asymptotically optimal estimate without assuming a specific data distribution.

Oracle Approximating Shrinkage (OAS) The OAS estimator [25] refines the Ledoit-Wolf approach by assuming the data follows a Gaussian distribution. This assumption allows for a more precise estimation of the shrinkage intensity, particularly for small sample sizes. The OAS shrinkage parameter ρ_{OAS} is given by:

$$\rho_{OAS} = \min\left(1, \frac{(1 - \frac{2}{d})\mathrm{tr}(S^2) + \mathrm{tr}^2(S)}{(n + 1 - \frac{2}{d})(\mathrm{tr}(S^2) - \frac{\mathrm{tr}^2(S)}{d})}\right) \tag{2.13}$$

OAS is specifically designed to minimize Mean Squared Error (MSE) under Gaussianity and often outperforms standard shrinkage when the data distribution is approximately normal.

2.2.4 Gaussianity Assumptions and Analysis

In the context of generative modeling for foundation feature spaces, the validity of the Bayes classifier depends on how well the class-conditional Gaussian distributions approximate the true data distribution. The following analytical framework establishes the statistical basis for these assumptions.

Univariate Analysis: Kurtosis and Skewness

The first layer of theoretical justification involves examining the univariate properties of individual feature components.

Kurtosis This statistic measures the "tailedness" and peakedness of a probability distribution relative to a normal distribution [3]. As illustrated in Figure 2.3, a perfectly Gaussian distribution has an excess kurtosis of zero (mesokurtic). Distributions with high kurtosis (leptokurtic, $Kurtosis > 0$) exhibit sharper peaks and heavier tails, indicating a higher concentration of data around the mean and a greater likelihood of extreme outliers. Conversely, low kurtosis (platykurtic, $Kurtosis < 0$) indicates a flatter peak and thinner tails. Empirical evidence suggests that for many foundation model embeddings, kurtosis values typically fall within the range of $[-1, 1]$, justifying the Gaussian approximation.

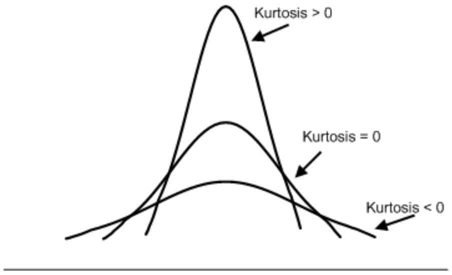


Figure 2.3: Visual representation of kurtosis levels: leptokurtic (> 0), mesokurtic (= 0) and platykurtic (< 0) distributions [3].

Skewness This measure quantifies the asymmetry of a distribution around its mean. As shown in Figure 2.4, a symmetric distribution (center) is a fundamental assumption of the Gaussian model, implying that the mean $\boldsymbol{\mu}_c$, median, and mode coincide, providing an adequate representation of the class prototype within the embedding space. A positive skew (right) indicates a distribution with a long tail extending toward higher values, while a negative skew (left) indicates a long tail extending toward lower values. Significant skewness would suggest that the class-conditional mean is no longer a central representation of the feature distribution.

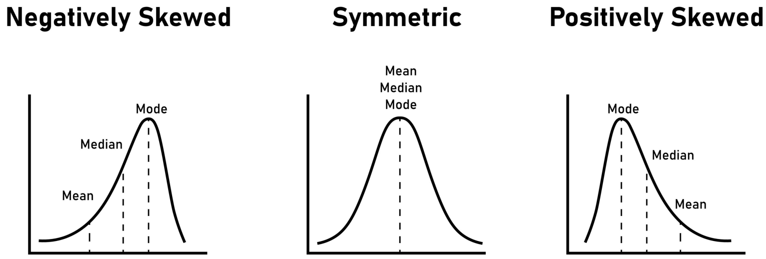


Figure 2.4: Comparison of negatively-skewed, symmetric and positively-skewed distributions [4].

Multivariate Analysis: Mahalanobis Distance

While univariate normality is a prerequisite, the Bayes classifier operates in a high-dimensional joint distribution. To assess the multivariate Gaussianity of the feature space, we utilize the Squared Mahalanobis Distance:

$$D^2(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \tag{2.14}$$

Theoretically, if a class-conditional distribution $p(\mathbf{x} \mid y = c)$ is a d-dimensional Gaussian, the squared Mahalanobis distance between the samples and the class mean follows a chi-square (χ^2) distribution with d degrees of freedom.

Degrees of Freedom and Rank The behavior of this distance is sensitive to the rank of $\mathbf{\Sigma}_c$. If the number of training samples N is smaller than the dimensionality d, the covariance matrix becomes rank-deficient, necessitating regularization (such as diagonal loading) to maintain numerical stability during the inversion process required for the MAP decision rule.

Correlation Modeling The multivariate assumption also addresses inter-feature dependencies. While a diagonal covariance model ($\text{diag}(\sigma^2_{c,1}, \ldots, \sigma^2_{c,d})$) assumes independence between features, a full covariance model attempts to capture the joint geometric structure of the class in the embedding space.

Spectral Decay and Dimensionality

The effectiveness of Gaussian modeling is further supported by the spectral decay of the feature space. By performing Principal Component Analysis (PCA) on the embeddings, one can observe the distribution of eigenvalues.

Information Concentration A steep decay in the leading eigenvalues indicates that most of the discriminative variance is concentrated within a lower-dimensional subspace.

Subspace Validity This concentration suggests that even though the nominal dimensionality d is high (e.g., $d = 768$), the classes occupy structured manifolds where Gaussian density estimation provides a strong and stable inductive bias.

2.3 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisition of knowledge from sequences of tasks or data streams over time. As illustrated in Figure 2.5, a **task** in this context is defined as a specific learning experience or dataset –such as Task A, B, or C. Each task is potentially characterized by unique data distributions or objectives, as symbolized by the varying shapes (e.g., circles vs. hearts) above the task labels.

Unlike conventional training approaches that rely on a fixed dataset and a stationary environment, CL emphasizes the arrival of these tasks at distinct intervals along the **time** axis. This sequential arrival requires the model to be inherently non-stationary, adapting its internal representations as environments evolve. The primary challenge in this paradigm lies in the ongoing integration of new information from a current task (e.g., Task B) while simultaneously preserving the knowledge previously acquired from Task A.

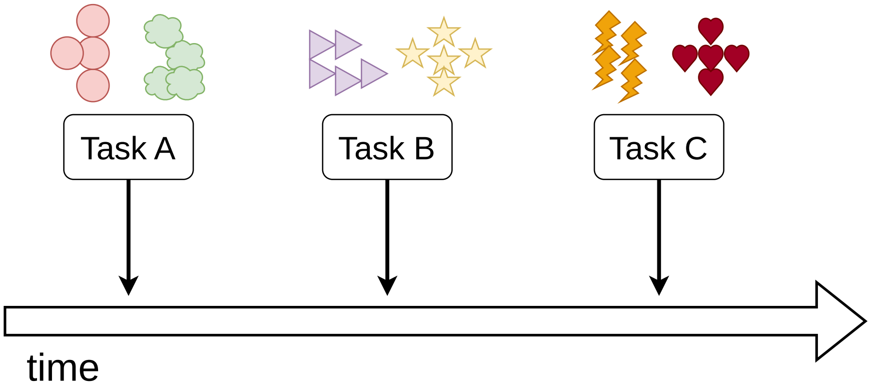


Figure 2.5: A sequential task arrival model for Continual Learning, depicting the transition from Task A to Task C over time, where each task represents a unique set of features or data distributions.

Without specific architectural or algorithmic safeguards, this sequential learning process inevitably triggers **catastrophic forgetting** [26]. This phenomenon occurs when the assimilation of new knowledge leads to the degradation or total loss of prior learning. As illustrated in Figure 2.6, the dilemma of forgetting becomes apparent when a *New Object Class* is introduced to an existing classifier, forcing a choice between two traditional but flawed strategies:

- **Fine-Tuning:** Updating the model using only the current task data is computationally efficient but leads directly to catastrophic forgetting [26]. Because the optimization process focuses solely on the new objective, the internal representations associated with previously learned classes are overwritten, causing the model to lose its original predictive capabilities.
- **Retraining from Scratch:** While joint training on the full historical dataset ($\mathcal{D}_{old} \cup \mathcal{D}_{new}$) technically prevents forgetting, it faces two significant hurdles. First is the issue of **Data Privacy and Availability**: in many real-world scenarios, the original data $\mathcal{D}_{old}$ is often deleted or inaccessible due to privacy regulations. Second is the **Computational Bottleneck**: as the system's lifespan increases, the cumulative dataset grows, making the energy and time required to retrain the model unsustainable in continuously evolving environments.

2.3.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. **Plasticity** refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. **Elasticity** (or stability), in contrast, emphasizes preserving the parameters and representations that are crucial for retaining old knowledge [9]. Achieving an appropriate equilibrium between these two properties is known as the **stability–plasticity dilemma** [26].

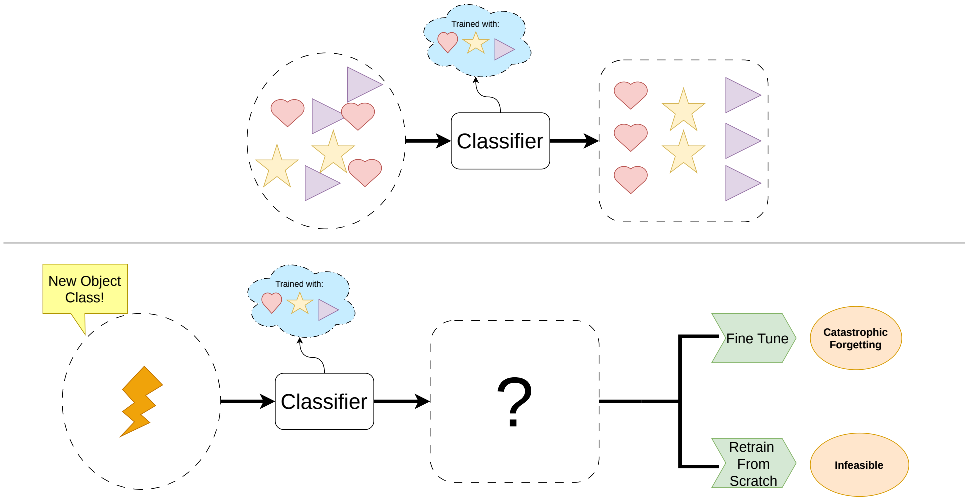

Trained with:
Classifier
New Object Class!
Trained with:
Classifier
?
Fine Tune
Catastrophic Forgetting
Retrain From Scratch
Infeasible

Figure 2.6: Standard Object Classifier comparison. The upper path shows standard inference, while the lower path illustrates the dilemma when a new class is introduced: the trade-off between catastrophic forgetting via fine-tuning and the infeasibility of retraining from scratch.

2.3.2 Main Paradigms

As shown in Figure 2.7, two principal paradigms define how continual learning problems are formulated. **Task-incremental learning (Task-IL)** assumes that the task identity is available during inference, allowing for task-specific components to be isolated [27]. **Class-incremental learning (Class-IL)**, on the other hand, does not provide the task identity at test time, requiring the model to distinguish between all classes seen across all tasks simultaneously [5].

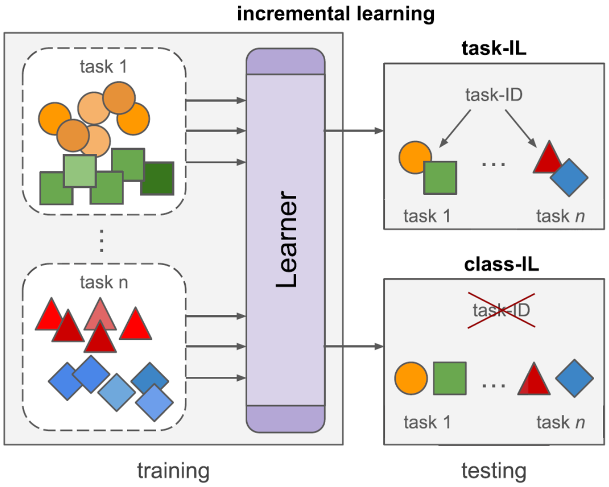

incremental learning
task 1
task n
Learner
task-IL
task-ID
task 1
task n
class-IL
task-ID
task 1
task n
training
testing

Figure 2.7: Class and Task Incremental Learning. Image taken from [5].

2.3.3 Mitigation Strategies

To address the stability-plasticity dilemma, CL strategies are generally categorized into four main families: rehearsal-based, regularization-based, architecture-based, and representation-based (specifically prompt-based) methods.

Rehearsal Methods

To preserve past performance, these methods reuse samples from previously learned tasks while training on the current task.

- **Experience Replay (ER)** [28]: Maintains a small episodic memory buffer containing a sample of old, representative examples to be interleaved with new data.
- **Average Gradient Episodic Memory (A-GEM)** [29]: Improves efficiency by using the buffer to constrain the direction of parameter updates, ensuring they do not interfere with past tasks.
- **Dark Experience Replay (DER)** [30]: Enhances ER by incorporating knowledge distillation on the logits of replayed examples to preserve the "dark knowledge" of the model.
- **Generative Replay** [31]: A non-data-storing method where a generator is trained to produce synthetic features or samples of previously learned classes, avoiding the privacy concerns of raw data storage.

Regularization Methods

These methods limit updates to parameters identified as important for prior learning by introducing penalty terms in the loss function.

- **Elastic Weight Consolidation (EWC)** [32]: Identifies critical parameters using the Fisher Information Matrix and restricts their movement during new task training.
- **Synaptic Intelligence (SI)** [33]: Calculates importance weights online by accumulating their contribution to the reduction of the loss.
- **Learning Without Forgetting (LwF)** [34]: A functional regularization method that uses knowledge distillation to ensure the model's outputs for new data remain consistent with the previous model's predictions.
- **Adaptive Plasticity Improvement (API)** [9]: Enforces an update direction orthogonal to the important subspaces of previous tasks using Dual Gradient Projection Memory (Dual-GPM).

Architecture-based Methods

These methods mitigate forgetting by modifying the network architecture to increase or reorganize its capacity.

- **Parameter Isolation**: Identifies and freezes parameters important for previous tasks –often through pruning or gradient masking– so only a subset of weights is updated for the current task [35].
- **Dynamic Expansion**: The architecture grows as new tasks arrive. For instance, [36] uses an expandable super-feature extractor followed by pruning, while [37] expands the network only when the current capacity is insufficient.

Representation and Prompt-based Methods

The emergence of foundation models (e.g., ViTs) has enabled parameter-efficient fine-tuning (PEFT) where the backbone remains frozen and learning is contained within small "prompts".

- **Learning to Prompt (L2P)** [38]: Uses a prompt pool where prompts are selected based on the similarity between input embeddings and learned keys.
- **Dual Prompt** [39]: Refines L2P by separating representations into fixed G-prompts (General) and task-specific E-prompts (Expert).
- **CODA-Prompt** [40]: Introduces a fully differentiable method where the prompt is a weighted sum of components dynamically calculated via attention mechanisms.
- **Advanced Prompting**: Recent methods like **InFLoRA** [41] and **DOLFIN** [42] apply low-rank adaptation and distributed learning, while **APT** [43] injects key-value tokens directly into transformer layers to reduce inference overhead.

Chapter 3

Methodology

3.1 Proposed Architecture

As shown in 3.1, our CIL framework is built upon a simple yet structurally motivated decomposition: (i) a frozen feature extractor based on a visual foundation model, and (ii) a probabilistic classification layer that models each object category through an independent Gaussian distribution in the embedding space. Given an input image, the feature extractor maps it to a fixed high-dimensional representation. For each class, we maintain a class-conditional multivariate Gaussian characterized by its mean vector and covariance matrix, estimated incrementally from observed samples. Classification is performed according to the maximum likelihood principle, assigning the input to the class whose Gaussian model yields the highest likelihood under the stationary embedding representation. This modular separation between representation learning and statistical modeling enables incremental updates without backpropagation or parameter adaptation: adding a new class requires only estimating its distribution in the fixed feature space, while previously learned classes remain unaffected at the representation level.

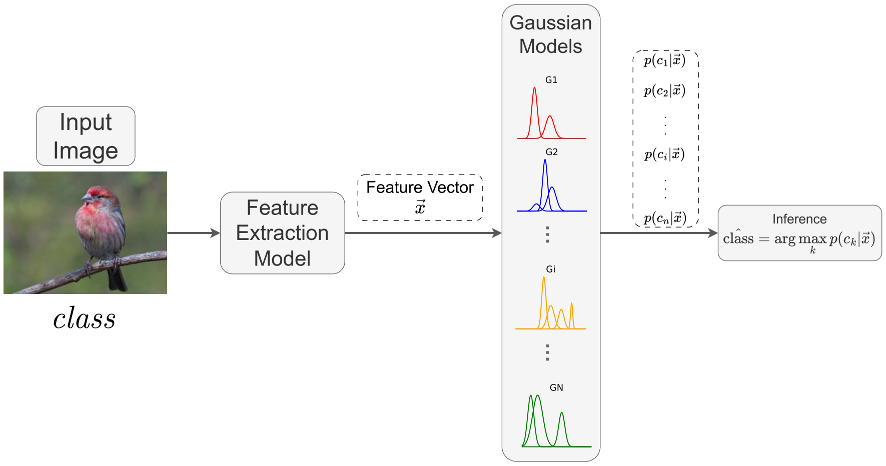

Input Image
class
Feature Extraction Model
Feature Vector $\vec{x}$
Gaussian Models
G1
G2
Gi
GN
$p(c_1|\vec{x})$
$p(c_2|\vec{x})$
$p(c_i|\vec{x})$
$p(c_n|\vec{x})$
Inference
$\hat{\text{class}} = \arg\max_k p(c_k|\vec{x})$

Figure 3.1: Block diagram of the proposed approach.

3.1.1 Datasets and Experimental Protocol

We follow the standard Continual Learning (CL) protocol where the model learns a sequence of disjoint tasks and calculates the average accuracy and the forgetting. The specific configurations for the datasets are as follows:

- **Split CIFAR-100:** This dataset is a widely utilized benchmark in the continual learning literature. It partitions the original CIFAR-100 [21] dataset into 10 disjoint tasks, with each task containing 10 distinct classes.
- **Split ImageNet-R:** This benchmark is constructed from ImageNet-R [20], a dataset containing renditions of ImageNet classes such as art, cartoons, and sketches. We randomly divide the 200 classes into 10 sequential tasks, with 20 classes per task, introducing a significant domain shift challenge.
- **Split Caltech-256:** Designed to evaluate the model on higher-resolution images with greater object variability, this benchmark utilises the Caltech-256 [22] dataset. The first 250 of the original 256 categories are used and sequentially divided into 10 disjoint tasks of 25 classes each; the remaining six categories are discarded so that the benchmark has a fixed, balanced number of classes per task. The resulting protocol contains approximately 28 100 images with an average of $\overline{n_c} \approx 112$ training samples per class, which makes the per-class budget the smallest of the three constituent datasets and stresses the high-dimensional covariance estimators introduced in Section 3.2.
- **Split Joint Dataset:** This benchmark serves as a comprehensive stress test by aggregating the 30 tasks from the three aforementioned datasets into a single, heterogeneous stream. The tasks are presented to the system in a randomized order, forcing the model to navigate extreme distribution shifts between natural, artistic, and high-resolution images. This long-horizon sequence is specifically designed to evaluate the model's resilience against catastrophic forgetting over an extended period of diverse learning objectives.

Regarding forgetting, the proposed framework does not exhibit forgetting in the conventional sense, as previously learned class statistics are never modified when new classes are introduced. The measured forgetting values therefore primarily reflect the increased classification difficulty as the number of competing classes grows, rather than degradation of previously learned representations.

3.2 Covariance Estimation

As mentioned, due to the high the dimensionality of the extracted feature, the empirical sample covariance matrix (SCM) is a poor estimator: its eigenvalues are biased, and the matrix is singular if $n < d$. Our benchmarks are tested using different convariance estimation methods:

3.2.1 Diagonal Covariances

In this configuration, we assume that the features within each class c are independent, leading to a diagonal covariance matrix $\Sigma_c = \text{diag}(\sigma_{c,1}^2, \sigma_{c,2}^2, \dots, \sigma_{c,d}^2)$. This simplification significantly reduces the number of parameters to estimate from $O(d^2)$ to $O(d)$, which is particularly advantageous when the dimensionality $d = 768$ is large relative to the number of samples per class.For each class $c \in \{1, \dots, C\}$, the mean vector μ_c and the diagonal elements of Σ_c are computed as:

$$\mu_c = \frac{1}{n_c} \sum_{i=1}^{n_c} x_i, \quad \sigma_{c,j}^2 = \frac{1}{n_c} \sum_{i=1}^{n_c} (x_{i,j} - \mu_{c,j})^2 + \varepsilon \tag{3.1}$$

where $\varepsilon = 10^{-6}$ is a small constant added to ensure numerical stability and strictly positive variances. The log-likelihood of a test sample $\mathbf{x}$ belonging to class c is determined by the discriminant function:

$$g_c(\mathbf{x}) = \ln P(\mathbf{x}|c) + \ln P(c) \tag{3.2}$$

Under the assumption of a uniform prior $P(c)$ and Gaussian distribution, this simplifies to:

$$g_c(\mathbf{x}) = -\frac{1}{2} \sum_{j=1}^{d} \left(\ln(2\pi\sigma_{c,j}^2) + \frac{(x_j - \mu_{c,j})^2}{\sigma_{c,j}^2} \right) \tag{3.3}$$

In the implementation, this is computed using a pre-calculated constant $\ln A_c = -\frac{d}{2}\ln(2\pi) - \frac{1}{2}\sum \ln(\sigma_{c,j}^2)$ and the squared Mahalanobis distance weighted by the inverse of the diagonal variances.

3.2.2 Full Covariances

To account for the dependencies between different dimensions of the feature vector, we estimate a full covariance matrix $\Sigma_c \in \mathbb{R}^{d \times d}$ for each class c. While this provides a richer representation of the class distribution, the estimation of d^2 parameters (where $d = 768$) is prone to ill-conditioning when the number of samples n_c is limited.The mean vector μ_c and the empirical covariance Σ_c are calculated as:

$$\mu_c = \frac{1}{n_c} \sum_{i=1}^{n_c} x_i, \quad \Sigma_c = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (x_i - \mu_c)(x_i - \mu_c)^T \tag{3.4}$$

To ensure the matrix is invertible and to improve numerical stability, a regularizatio is applied by adding a small identity term to the diagonal: $\hat{\Sigma}_c = \Sigma_c + \varepsilon I$, with $\varepsilon = 10^{-6}$.The implementation performs the inversion and determinant calculation using the eigendecomposition of the symmetric covariance matrix, $\hat{\Sigma}_c = P\Lambda P^T$, where Λ contains the eigenvalues λ_j. To further ensure stability, eigenvalues are clipped to a minimum value of 10^{-6}. The inverse and the log-determinant are then computed as:

$$\hat{\Sigma}_c^{-1} = P\Lambda^{-1}P^T, \quad \ln|\hat{\Sigma}_c| = \sum_{j=1}^{d} \ln(\lambda_j) \tag{3.5}$$

The discriminant function for classification is the full quadratic form:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \hat{\Sigma}_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.6}$$

where $\ln A_c = -\frac{1}{2}(d\ln(2\pi) + \ln|\hat{\Sigma}_c|)$.

3.2.3 Hybrid Covariance

The hybrid covariance estimator is designed to leverage the correlation information for distance metrics while mitigating the numerical instability inherent in calculating the determinant of high-dimensional, potentially rank-deficient matrices. In this approach, we decoupling the estimation of the Mahalanobis distance and the normalization constant.For each class c, the full covariance matrix $\hat{\Sigma}_c$ is estimated and regularized as described in the previous section:

$$\hat{\Sigma}_c = \Sigma_c + \varepsilon I \tag{3.7}$$

The quadratic form of the log-likelihood (the Mahalanobis distance) uses the full inverse $\hat{\Sigma}_c^{-1}$, calculated via eigendecomposition to capture the feature interactions. However, the log-determinant—which represents the "volume" of the distribution—is computed using only the diagonal elements of the covariance matrix (the variances):

$$\ln|\Sigma_c|_{\text{diag}} = \sum_{j=1}^{d} \ln(\sigma_{c,j}^2) \tag{3.8}$$

where $\sigma_{c,j}^2$ are the diagonal entries of $\hat{\Sigma}_c$. The resulting discriminant function is defined as:

$$g_c(\mathbf{x}) = -\frac{1}{2}\left(d\ln(2\pi) + \sum_{j=1}^{d} \ln(\sigma_{c,j}^2) + (\mathbf{x} - \boldsymbol{\mu}_c)^T \hat{\Sigma}_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c)\right) \tag{3.9}$$

This hybrid formulation prevents the "vanishing determinant" problem that often occurs in high dimensions, where very small eigenvalues (even after regularization) can disproportionately inflate the log-likelihood and lead to biased classification toward classes with lower variance.

3.2.4 OAS Covariance

Oracle Approximating Shrinkage (OAS) is a shrinkage technique that computes an structurally optimized covariance matrix by finding an optimal compromise between the sample covariance matrix and a structured target (the identity matrix scaled by the average trace).The OAS estimator provides a more accurate estimation of the covariance Σ_c by minimizing the Mean Squared Error (MSE) under the assumption of Gaussianity. The estimated covariance is defined as:

$$\hat{\Sigma}_{\text{OAS},c} = (1 - \rho_c)\hat{\Sigma}_{\text{SCM},c} + \rho_c \hat{F}_c \tag{3.10}$$

where ρ_c is the shrinkage coefficient calculated specifically for class c, $\hat{\Sigma}_{\text{SCM},c}$ is the sample covariance matrix, and $\hat{F}_c = \frac{\text{tr}(\hat{\Sigma}_{\text{SCM},c})}{d} I$ is the shrinkage target.In our implementation, we utilize the scikit-learn `OAS` object to compute the precision matrix $\Psi_c = \hat{\Sigma}^{-1}_{\text{OAS},c}$ directly. This precision matrix is then used to calculate the Mahalanobis distance in the log-likelihood discriminant function:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Psi_c (\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.11}$$

where $\ln A_c$ is the normalization constant derived from the log-determinant of the OAS-estimated covariance matrix, calculated via the `slogdet` function for numerical stability. This approach ensures that the model captures class-specific correlations while remaining robust to the "curse of dimensionality."

3.2.5 Ledoit-Wolff

The Ledoit-Wolf Diagonal Anchor estimator addresses the instability of the sample covariance matrix S by shrinking it toward a target matrix T.

The selection of a target matrix T represents a prior assumption regarding the feature distribution. Both methods stabilize the estimator by pulling the noisy off-diagonal elements of the sample covariance matrix S toward zero, but they differ in how they treat the diagonal elements (the variances).

Diagonal Target

The diagonal anchor is defined as $T_{\text{diag}} = \text{diag}(s_{11}, s_{22}, \ldots, s_{dd})$. This approach assumes that the features are independent but allows for heteroscedasticity, meaning each dimension can have a different variance. Shrinkage toward this target preserves the scale of each individual feature as observed in the sample data while suppressing only the correlations. This is particularly useful when certain dimensions of the DINOv2 embeddings are naturally more informative or volatile than others.

Identity Target

The identity anchor is more restrictive, defined as $T_{\text{ident}} = \mu I$, where $\mu = \frac{1}{d}\text{tr}(S)$ is the average variance across all dimensions. This target assumes that the features are not only independent but also share a common variance (homoscedasticity). The resulting distribution is spherical. Because the embeddings used in this work are L2-normalized, the variances across dimensions tend to be similar, making the identity anchor a highly robust regularizer that prevents any single dimension from disproportionately influencing the distance metric.

Structural Summary

The fundamental difference between these two estimators lies in the flexibility of the variance components:

- **Diagonal Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = s_{ii}$. It yields an axis-aligned ellipsoidal distribution.
- **Identity Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = (1-\gamma)s_{ii} + \gamma\mu$. It yields a distribution that is pulled toward a perfectly spherical geometry.

3.2.6 Shared Covariance

The Shared Covariance framework assumes that the internal noise or feature variability is homoscedastic across all classes. In this paradigm, all classes share a single covariance matrix Σ, which captures the global structure of the feature space. This significantly reduces the total number of parameters to be estimated compared to class-specific models. Depending on the constraints placed on Σ, we evaluate three distinct variations:

Shared Diagonal Covariance

By enforcing a global diagonal covariance matrix $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_d^2)$, we simplify the model to assume that feature dimensions are independent. The shared variance for each feature j is estimated by pooling the squared residuals from all C classes:

$$\sigma_j^2 = \frac{1}{N}\sum_{c=1}^{C}\sum_{i=1}^{n_c}(x_{i,j} - \mu_{c,j})^2 + \varepsilon \tag{3.12}$$

where N is the total number of training samples and $\mu_{c,j}$ is the mean of class c for feature j. The classification of a test sample $\mathbf{x}$ reduces to finding the class c that minimizes the weighted Euclidean distance:

$$c^* = \arg\min_c \sum_{j=1}^{d} \frac{(x_j - \mu_{c,j})^2}{\sigma_j^2} \tag{3.13}$$

Shared Full Covariance

In the shared full covariance model (equivalent to the generative assumption of Linear Discriminant Analysis), we estimate a dense matrix $\Sigma \in \mathbb{R}^{d \times d}$ that captures the correlations between all feature dimensions. To ensure numerical stability in high-dimensional spaces, we utilize the Oracle Approximating Shrinkage (OAS) estimator to compute the pooled covariance. The classification rule utilizes the shared precision matrix $\Psi = \Sigma^{-1}$ to minimize the full squared Mahalanobis distance:

$$c^* = \arg\min_c (\mathbf{x} - \boldsymbol{\mu}_c)^T \Psi (\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.14}$$

Shared Hybrid Covariance

The shared hybrid approach serves as a regularized middle ground. It utilizes the full shared precision matrix Ψ for the quadratic distance calculation to capture feature interactions, but simplifies the normalization constant (the log-determinant) by considering only the diagonal elements of the shared matrix. This prevents the classification scores from being dominated by extreme eigenvalues while still benefiting from the descriptive power of the full correlation structure.

3.2.7 Mixture of Gaussians

In order to capture potential multi-modality within the feature space of a single class, we employ Gaussian Mixture Models (GMM). We implemented the following models using the previously defined diagonal, full and hybrid covariances approaches.

Mixture of 2 Gaussians

Forcing each class to be defined as a GMM with $K = 2$ components, each class c can be modeled by a probability density function:

$$p(\mathbf{x}|c) = \sum_{k=1}^{K} \pi_{c,k} \mathcal{N}(x|\mu_{c,k}, \Sigma_{c,k}) \tag{3.15}$$

where $\pi_{c,k}$ represents the mixing weight of the k-th component for class c, subject to $\sum \pi_{c,k} = 1$. In this specific variant, the covariance matrices $\Sigma_{c,k}$ are restricted to be diagonal to maintain computational efficiency and prevent over-fitting in the high-dimensional DINOv2 space.The parameters $\{\pi_{c,k}, \mu_{c,k}, \Sigma_{c,k}\}$ are estimated using the Expectation-Maximization (EM) algorithm. To ensure numerical stability, a small constant $\varepsilon = 10^{-5}$ is added to the diagonal of each covariance matrix. The log-likelihood for a test sample $\mathbf{x}$ is calculated using the *LogSumExp* operator to avoid underflow:

$$\ln p(\mathbf{x}|c) = \mathrm{lnaddexp}\left(\ln(\pi_{c,1}\mathcal{N}(x|\mu_{c,1}, \Sigma_{c,1})), \ln(\pi_{c,2}\mathcal{N}(x|\mu_{c,2}, \Sigma_{c,2}))\right) \tag{3.16}$$

In the implementation, this is expanded as:

$$\ln p(\mathbf{x}|c) = \ln\left(\exp(\ln \pi_{c,1} + \ln A_{c,1} - \frac{1}{2}d_{M,1}) + \exp(\ln \pi_{c,2} + \ln A_{c,2} - \frac{1}{2}d_{M,2})\right) \tag{3.17}$$

where $d_{M,k}$ is the diagonal Mahalanobis distance to the k-th component and $\ln A_{c,k}$ is the normalization constant. This approach allows the model to better represent classes that may contain significantly different visual sub-types or clusters within the latent embedding space.

Mixture of up-to-4 Gaussians

To adapt to the varying complexity of different classes within the different datasets, we implement a flexible Gaussian Mixture Model where the number of components K_c for each class c is not fixed. Instead, the model selects the most appropriate number of components $K_c \in \{1, 2, 3, 4\}$ by minimizing the Bayesian Information Criterion (BIC).The BIC is defined as:

$$\mathrm{BIC} = \ln(n)k - 2\ln(\hat{L}) \tag{3.18}$$

where n is the number of samples in the class, k is the number of free parameters to be estimated, and $\hat{L}$ is the maximized value of the likelihood function. By using diagonal covariance matrices, the number of parameters per component is kept low $(2d+1)$, allowing the selection process to favor multi-modal distributions only when the increase in likelihood justifies the additional parameters. For each component k in class c, the diagonal covariance $\Sigma_{c,k} = \mathrm{diag}(\sigma^2_{c,k,1}, \ldots, \sigma^2_{c,k,d})$ is regularized with $\varepsilon = 10^{-6}$. The classification of a test sample $\mathbf{x}$ involves calculating the weighted log-likelihood across all active classes:

$$\ln p(\mathbf{x}|c) = \mathrm{LSE}_{k=1}^{K_c}\left(\ln \pi_{c,k} - \frac{1}{2}\left(\sum_{j=1}^{d} \ln(2\pi\sigma^2_{c,k,j}) + \sum_{j=1}^{d} \frac{(x_j - \mu_{c,k,j})^2}{\sigma^2_{c,k,j}}\right)\right) \tag{3.19}$$

where LSE denotes the LogSumExp operator. This methodology allows for a more granular representation of class distributions—using a single Gaussian for simple, compact classes and up to four Gaussians for classes with high intra-class variance or distinct visual sub-clusters—while preserving the numerical efficiency of diagonal estimators.

3.3 Probabilistic Classification in High Dimension

In high-dimensional feature spaces, such as those generated by foundational vision models, probabilistic classification faces significant challenges regarding calibration. While the Maximum A Posteriori (MAP) decision rule identifies the most likely class, the resulting posterior probabilities $p(y = c \mid \mathbf{x})$ often exhibit extreme overconfidence. This phenomenon is characterized by the model assigning near-unit probability to predictions, even when they are incorrect, effectively masking the true uncertainty of the classification.

The Log-Gap and Uncertainty Collapse We model the distribution of features $\mathbf{x} \in \mathbb{R}^d$ using a Gaussian generative framework. Under the assumption of diagonal covariance matrices $\mathbf{\Sigma}_c = \operatorname{diag}(\sigma_{c,1}^2, \ldots, \sigma_{c,d}^2)$, the log-likelihood for a class c is defined by:

$$\log p(\mathbf{x} \mid y = c) = \ln \mathcal{A}_c - \frac{1}{2} \sum_{i=1}^{d} \frac{(x_i - \mu_{c,i})^2}{\sigma_{c,i}^2} \tag{3.20}$$

where $\ln \mathcal{A}_c = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \sum \ln(\sigma_{c,i}^2)$ represents the normalization constant. In high dimensions ($d \gg 1$), the log-likelihood is dominated by the sum of squared standardized residuals.A critical metric for calibration is the Log-Gap (G), defined as the difference between the log-likelihood of the most probable class (c_1) and the runner-up (c_2):

$$G = \log p(\mathbf{x} \mid y = c_1) - \log p(\mathbf{x} \mid y = c_2) \tag{3.21}$$

As d increases, G tends to scale linearly, often reaching values where the ratio of likelihoods is orders of magnitude apart. When these likelihoods are converted to a posterior distribution, the model's entropy collapses, rendering the probability mass insensitive to the underlying class overlap.

Bayesian Temperature Scaling To recover reliable uncertainty estimates without altering the MAP decision boundary, we introduce a Temperature Scaling parameter T directly into the Bayesian inference process. In this generative context, we treat T as a smoothing factor applied to the log-likelihoods before marginalization.The calibrated posterior probability is computed by normalizing the scaled likelihoods over the set of active classes $\mathcal{Y}$:

$$p(y = c \mid \mathbf{x}; T) = \exp\left(\frac{1}{T} \log p(\mathbf{x} \mid y = c) - \mathcal{L}_E\right) \tag{3.22}$$

where $\mathcal{L}_E$ is the log-evidence (or log-marginal likelihood) calculated using the logsumexp operator to ensure numerical stability:

$$\mathcal{L}_E = \operatorname{logsumexp}_{j \in \mathcal{Y}} \left(\frac{1}{T} \log p(\mathbf{x} \mid y = j)\right) \tag{3.23}$$

By choosing $T > 1$, we contract the Log-Gap, effectively "softening" the distribution. This methodological adjustment ensures that the model's predictive confidence—and its resulting entropy $\mathcal{H}(p)$—better reflects the task's difficulty and the inherent density of the feature manifold, specifically aiming to minimize the Expected Calibration Error (ECE).

Calibration Metrics The quality of the probabilistic output is evaluated using the Expected Calibration Error (ECE), which partitions the predictions into M bins and computes the weighted average of the absolute difference between accuracy and confidence:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \tag{3.24}$$

A well-calibrated system achieves an ECE ≈ 0, ensuring that the predicted probability serves as a reliable proxy for the true likelihood of correctness, a critical requirement for safety and interpretability in industrial computer vision applications.

3.4 Generative-Discriminative Fusion and Re-ranking Strategies

In high-dimensional classification tasks, relying on a single metric often fails to capture the full complexity of the feature space. While generative approaches (Mahalanobis-based) model the underlying class-conditional distributions, discriminative approaches (Cosine-based) excel at capturing angular separability on the unit hypersphere. To leverage the strengths of both, we evaluate several fusion and re-ranking strategies.

Importantly, these strategies are agnostic to the specific structure of the class-conditional distributions; they can be applied whether the underlying generative model utilizes a **diagonal covariance** matrix for computational efficiency or a **full covariance** matrix to capture complex feature correlations.

3.4.1 HardGate: Hierarchical Task Gating

This strategy implements a strict two-step decision process. First, a global gating mechanism identifies the most probable task t^* for a given input $\mathbf{x}$. Once the task is fixed, the final classification is restricted solely to the classes belonging to that task.The task-level distribution is modeled as a Gaussian $\mathcal{N}(\mu_t, \Sigma_t)$, where the parameters are estimated by pooling all samples from the classes belonging to task t. The inference logic is:

1. **Task Selection:** $t^* = \arg\max_t \left(\ln A_t - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^T \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)\right)$
2. **Local Classification:** $c^* = \arg\max_{c \in \text{Task}_{t^*}} \left(\ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c)\right)$

This approach significantly reduces the search space during inference but is highly sensitive to errors in the initial gating step (Task-IL vs. Class-IL interference).

3.4.2 Top5-TaskReRank: Top-5 Candidate Re-ranking via Task Likelihood

Unlike the strict gating, this strategy performs an initial global search across all active classes but uses task-level information to refine the final choice among the most likely candidates.The process follows these steps:

1. **Candidate Selection:** The model identifies the Top-5 classes with the highest log-likelihoods across the entire set of active classes.
2. **Task-Guided Verification:** For each candidate in the Top-5, the model retrieves the likelihood of the parent task $p(\mathbf{x}|t)$.
3. **Selection:** The final class c^* is chosen as the one among the Top-5 candidates whose parent task yields the highest task-level likelihood.

This strategy provides a "safety net" where the fine-grained class similarity is cross-referenced with the coarser task-level distribution to resolve ambiguities.

3.4.3 TopK-TaskFilter: Top-K Task Candidate Filtering

This strategy is a hybrid between strict gating and global search. It expands the search space of Gating by considering the N most probable tasks (where $N = 3$ in the implementation) and evaluating all classes within those selected tasks.The decision rule is formulated as:

$$c^* = \arg \max_{c \in \bigcup \text{Top-}N(\text{Tasks})} \left(\ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \right) \tag{3.25}$$

By evaluating a "neighborhood" of likely tasks, this method mitigates the risk of a single incorrect gating decision while maintaining computational efficiency by ignoring tasks that are highly unlikely according to the pooled task-level Gaussian models.

3.4.4 SoftTaskFusion: Soft Class–Task Log-Probability Fusion

The Soft Task-Aware Fusion strategy mitigates the rigidity of hierarchical gating by treating the task assignment as a latent variable in a joint probability framework. In this approach, the final decision score for a class c is determined by combining its class-conditional log-likelihood with the log-likelihood of its parent task $t(c)$. Mathematically, given a feature vector $\mathbf{x}$, the final score S_c for each active class is defined as:

$$S_c(\mathbf{x}) = \ln P(\mathbf{x}|c) + \ln P(\mathbf{x}|t(c)) \tag{3.26}$$

where $\ln P(\mathbf{x}|c)$ and $\ln P(\mathbf{x}|t(c))$ represent the Gaussian log-likelihoods at the class and task levels, respectively. These are computed as:

$$\ln P(\mathbf{x}|i) = \ln A_i - \frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i), \quad i \in c, t \tag{3.27}$$

During inference, the task-level log-likelihoods $\ln P(\mathbf{x}|t)$ are calculated for all tasks currently seen in the stream. These values are then “expanded” or broadcasted to all constituent classes within those tasks. The classification rule is given by:

$$c^* = \arg \max_{c \in \mathcal{C}_{active}} [S_c(\mathbf{x})] \tag{3.28}$$

By performing an additive fusion in the log-domain (equivalent to a product of probabilities), this method allows the model to “boost” the scores of classes whose parent task distribution highly matches the input, while still allowing a class to be selected even if its task-level likelihood is not the absolute maximum, provided the class-level evidence is sufficiently strong. This approach effectively balances global class competition with coarse task-level guidance.

3.4.5 TaskMeanShrink: Task-Level MAP Mean Shrinkage

To improve the robustness of the class-level generative models, we implement a Maximum A Posteriori (MAP) estimation for the mean vectors. This strategy, often referred to as Mean Shrinkage, addresses the high variance in class-specific estimates—especially when data is scarce—by anchoring them to a task-level prior.

In this framework, each class c belonging to a task t is modeled using a task-aware mean $\mu_{c,\text{MAP}}$. This mean is a convex combination of the empirical class mean $\hat{\mu}_c$ and the pooled task mean μ_t:

$$\mu_{c,\text{MAP}} = \frac{n_c \hat{\mu}_c + \lambda \mu_t}{n_c + \lambda} \tag{3.29}$$

where n_c is the number of samples in class c, and λ is a shrinkage hyperparameter (set to 10.0 in our experiments) that controls the influence of the task-level prior. The task mean μ_t is computed by pooling the features of all classes within task t:

$$\mu_t = \frac{1}{\sum_{i \in \text{Task}_t} n_i} \sum_{i \in \text{Task}_t} \sum_{j=1}^{n_i} x_{i,j} \tag{3.30}$$

By shrinking the class means toward the task centroid, the model reduces over-fitting to class-specific noise while emphasizing features that characterize the broader task domain. Once the MAP means are calculated, they are used to compute the Mahalanobis distance in the log-likelihood score:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{c,\text{MAP}})^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_{c,\text{MAP}}) \tag{3.31}$$

This approach provides a “Bayesian” regularization effect, ensuring that classes with very few samples or high intra-class variance are pull towards a more stable task-level representation, thereby improving generalizability in the global classification space.

3.4.6 Top5-CosineReRank: Generative–Angular Cosine Re-ranking

The Generative-Angular Re-ranking strategy leverages both the statistical modeling of Mahalanobis-based distances and the angular separability of Cosine similarity. In high-dimensional latent spaces, particularly those generated by models trained with contrastive or foundation objectives like DINOv2, the direction of the feature vector often carries significant discriminative information that the Euclidean-based Mahalanobis distance might under-prioritize.

The inference procedure is structured as a two-stage pipeline:

1. **Generative Candidate Filtering:** For an input feature $\mathbf{x}$, we first compute the log-likelihood scores across all active classes $\mathcal{C}_{active}$ using the Gaussian generative model (utilizing either diagonal or full covariance matrices):

   $$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.32}$$

   From these scores, a candidate set $\mathcal{K}$ containing the Top-5 classes with the highest likelihood is extracted.

2. **Angular Re-ranking:** Within the subset $\mathcal{K}$, the final prediction c^* is determined by maximizing the Cosine similarity between the input feature $\mathbf{x}$ and the class mean vectors μ_c:

   $$c^* = \arg\max_{c \in \mathcal{K}} \frac{\mathbf{x} \cdot \boldsymbol{\mu}_c}{\|\mathbf{x}\| \, \|\boldsymbol{\mu}_c\|} \tag{3.33}$$

By using the Gaussian model as a broad filter and Cosine similarity as a fine-grained refiner, this approach ensures that the chosen class is not only statistically probable given the distribution volume but also geometrically aligned with the class centroid on the unit hypersphere. This hybrid approach is particularly effective in mitigating the “over-confidence” of Gaussian models in regions where class distributions overlap significantly.

3.4.7 HardGate-LDA: Hierarchical LDA Subspace Inference

This strategy combines global task gating with local expert models optimized via Linear Discriminant Analysis (LDA). By projecting high-dimensional DINOv2 features ($d = 768$) into a task-specific 9-dimensional subspace, the model maximizes the separability of classes within that task while reducing noise.

The process is defined as follows:

1. **Subspace Projection:** For each task t, we compute the within-class scatter matrix S_w and the between-class scatter matrix S_b using the classes belonging to that task. We solve the generalized eigenvalue problem $S_b v = \lambda S_w v$ to obtain the projection matrix $W_t \in \mathbb{R}^{768 \times 9}$, which contains the top 9 eigenvectors.

2. **Stage 1 (Hard Gating):** A test sample $\mathbf{x}$ is first assigned to a task t^* by maximizing the log-likelihood of task-pooled Gaussian distributions in the original 768D space:

   $$t^* = \arg\max_t \left(\ln A_t - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^T \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t) \right) \tag{3.34}$$

3. **Stage 2 (Local Subspace Classification):** The feature is projected into the specific subspace of task t^*: $\mathbf{x}' = W_{t^*}^T \mathbf{x}$. The final class is predicted using non-separable (full) covariance Gaussians modeled within this 9D space:

$$c^* = \arg \max_{c \in \text{Task}_{t^*}} \left(\ln A_{c,\text{proj}} - \frac{1}{2}(z' - \mu_{c,\text{proj}})^T \Sigma_{c,\text{proj}}^{-1}(z' - \mu_{c,\text{proj}}) \right) \tag{3.35}$$

3.4.8 SoftGate-LDA: Soft Hierarchical LDA Fusion

This other strategy improves upon the previous one by replacing the hard gating decision with a joint probabilistic fusion. This prevents a single error in the task selection stage from irrecoverably leading to a misclassification.

Instead of selecting t^* and then calculating the class likelihood, SoftGate-LDA evaluates a total score for every class across all tasks. The decision rule combines the global gating evidence with the local subspace likelihood:

$$c^* = \arg \max_{c \in \mathcal{C}_{\text{active}}} \left(\text{Score}_{\text{gating}}(t(c)) + \text{Score}_{\text{LDA}}(c | W_{t(c)}) \right) \tag{3.36}$$

where $t(c)$ is the task to which class c belongs. For every class, the feature $\mathbf{x}$ is projected into the subspace associated with its task. The scores are computed as:

- **Score$_{\text{gating}}$:** The log-likelihood of the parent task in the original 768D space.
- **Score$_{\text{LDA}}$:** The log-likelihood of the class in its task-specific 9D LDA subspace.

This additive fusion in the log-domain (equivalent to a product of probabilities) allows the local expert models to influence the task selection. If a class in Task A has an extremely high likelihood in its 9D subspace, it can override a slightly higher task-level likelihood for Task B, leading to more robust global classification.

3.4.9 OptFusion-MinMax: Optimised Min–Max Score Fusion

The Optimized Score Fusion strategy seeks to integrate the probabilistic rigor of generative Gaussian modeling with the angular robustness of Cosine similarity. Because log-likelihoods from a high-dimensional Gaussian distribution and Cosine similarity scores operate on vastly different numerical scales and ranges, a direct combination is mathematically unsound. To address this, we implement a dynamic fusion pipeline utilizing Min-Max Normalization and automated hyperparameter optimization.

For a given input $\mathbf{x}$, we first compute two independent score vectors across all active classes:

1. **Generative Score (S_{gen}):** The Mahalanobis-based log-likelihood $g_c(\mathbf{x})$.

2. **Discriminative Score (S_{dis}):** The Cosine similarity between x and the class mean μ_c:

$$\cos(\theta_c) = \frac{\mathbf{x} \cdot \boldsymbol{\mu}_c}{\|\mathbf{x}\| \, \|\boldsymbol{\mu}_c\|} \tag{3.37}$$

To perform the fusion, both score vectors are independently mapped to the range $[0, 1]$ using a Min-Max normalization function:

$$\text{Norm}(S) = \frac{S - \min(S)}{\max(S) - \min(S) + \varepsilon} \tag{3.38}$$

The final decision score S_{final} is then computed as a convex combination of the normalized vectors:

$$S_{final} = \alpha \cdot \text{Norm}(S_{gen}) + (1 - \alpha) \cdot \text{Norm}(S_{dis}) \tag{3.39}$$

The fusion weight $\alpha \in [0, 1]$ is determined using the Optuna framework, which performs a Bayesian optimization search over a validation subset (20% of the training data). By maximizing the Average Accuracy on this validation set, the model identifies the optimal balance between the generative and discriminative evidence before performing the final evaluation on the held-out test set. This approach ensures that the re-ranking is not merely a heuristic but is tuned to the specific geometric properties of the DINOv2 feature space.

3.4.10 TempScale: Generative Calibration via Temperature Scaling

The Temperature-Scaled Calibration strategy aims to adjust the "sharpness" of the generative probability distributions. In high-dimensional latent spaces, log-likelihood values can reach extreme magnitudes, potentially leading to over-confident or miscalibrated predictions when competing against a growing number of classes in a continual learning stream.

This approach introduces a global scalar parameter $T > 0$, known as the temperature, which scales the log-likelihood scores before the final decision is made. For a given input feature $\mathbf{x}$, the calibrated score S_c for class c is defined as:

$$S_c(\mathbf{x}; T) = \frac{1}{T} \left(\ln A_c - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \right) \tag{3.40}$$

The effect of the temperature parameter varies depending on its value:

- $T > 1$ **(Smoothing):** Reduces the difference between scores, softening the distribution and making the model less confident.
- $T < 1$ **(Sharpening):** Exaggerates the differences between scores, making the distribution more peaky around the most likely class.

To avoid data leakage and over-fitting, the optimal temperature T^* is determined through an automated search using the Optuna framework. The training dataset is split into a

"Statistics Set" (80%) used to estimate the Gaussian parameters (μ, Σ) and a "Validation Set" (20%) used to evaluate the temperature performance. The search is conducted across a logarithmic range (0.01 to 10.0) to find the T that maximizes the Average Accuracy on the validation set. Once optimized, the fixed T^* is applied to the final evaluation on the held-out test set, ensuring a calibrated balance across all tasks in the sequence.

3.5 Backbones Comparison

Throughout this work, DINOv2 [23] is used as the default backbone feature extractor for the Gaussian-based classifiers because of its self-supervised training regime and the strong empirical evidence of approximately Gaussian, well-conditioned class-conditional distributions presented in Section 3.7. However, the proposed framework is agnostic to the specific feature extractor used: any frozen encoder that produces fixed-dimensional embeddings can plug in unchanged into the downstream Gaussian classifier. To quantify the dependence of the framework on the choice of backbone, we further analyse the impact of the backbone architecture by re-running the Diag-Cov and Full-Cov variants using two supervised Vision Transformers from the `timm` library [44]: **ViT-base/16** (ViT-b, $d = 768$) and **ViT-medium/16** (ViT-m, $d = 512$), both pre-trained on ImageNet-21k [19]. The covariance estimation, hyper-parameters and evaluation protocol are kept identical across backbones to isolate the effect of the upstream representation; the results are reported in Section 4.5.

3.6 Model Metrics

To quantify performance, SOTA works [38] utilize two standard metrics derived from the test accuracy scores: the average accuracy and the average forgetting.

On the one hand, **Average Accuracy (A_t)**, presented on equation (3.41), measures overall performance after task t. It is the mean test accuracy across all tasks learned up to that point.

$$A_t = \frac{1}{t} \sum_{\tau=1}^{t} S_{t,\tau} \tag{3.41}$$

On the other hand, **Forgetting (F_t)**, presented on equation (3.42), quantifies how much the model has forgotten earlier tasks. It computes, for each past task, how much accuracy has dropped compared to its best historical score, and averages those drops.

$$F_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \left(\max_{k \le t-1} S_{k,\tau} - S_{t,\tau} \right) \tag{3.42}$$

3.6.1 Confidence and Calibration

While accuracy and forgetting measure the correctness of predictions, they do not account for the model's reliability or its tendency to over-confide in incorrect decisions—a common issue in Class-Incremental Learning. To address this, we introduce metrics that evaluate the alignment between the model's subjective probabilities and its actual performance.

Expected Calibration Error (ECE)

This metric quantifies the mismatch between predictive confidence and empirical accuracy. Following your implementation, we partition the predictions into M equally spaced bins based on their confidence scores. The ECE is defined as the weighted average of the absolute difference between the accuracy and confidence of each bin:

$$ECE_t = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| acc(B_m) - conf(B_m) \right| \tag{3.43}$$

where N is the total number of samples, B_m is the set of indices of samples whose prediction confidence falls into the m-th interval, and $acc(B_m)$ and $conf(B_m)$ represent the mean accuracy and mean confidence within that bin, respectively.

Confidence on Errors (CoE)

To specifically analyze the model's behavior when it fails, we measure the average confidence assigned to the winning (but incorrect) class. High CoE indicates a model that is "dangerously wrong", which is a critical factor in safety-oriented applications. It is computed for a task t as:

$$CoE_t = \frac{1}{|\mathcal{E}_t|} \sum_{i \in \mathcal{E}_t} \max(\hat{p}_i) \tag{3.44}$$

where $\mathcal{E}_t$ is the set of misclassified samples and $\hat{p}_i$ is the predicted probability distribution for sample i.

Average Predictive Entropy (H)

We utilize Shannon entropy to measure the uncertainty of the model's predictions. A higher entropy suggests that the model is distributing its probability mass across multiple classes, indicating lower certainty. For task t, the average entropy is calculated as:

$$H_t = \frac{1}{N}\sum_{i=1}^{N}\left(-\sum_{c=1}^{C} p_{i,c}\log(p_{i,c})\right) \tag{3.45}$$

where $p_{i,c}$ is the probability assigned to class c for sample i. Collectively, these metrics provide a multi-dimensional view of the model's stability and trustworthiness throughout the incremental learning process.

3.7 Gaussianity Analysis

Before committing to a class-conditional Gaussian classifier we must verify that the assumption is empirically defensible on the DINOv2 embedding space. Following the univariate moment tests introduced in Section 2.4 of the theoretical framework, we compute the sample skewness and excess kurtosis along every one of the $d = 768$ coordinates and for every class c in each of the three benchmark datasets (Split CIFAR-100, Split ImageNet-R, Split Caltech-256). For a class with n_c training samples and coordinate j, the moments are estimated as

$$\widehat{\mathrm{Skew}}_{c,j} \;=\; \frac{\frac{1}{n_c}\sum_{i=1}^{n_c}(x_{i,j}-\mu_{c,j})^3}{\hat{\sigma}_{c,j}^3}, \qquad \widehat{\mathrm{Kurt}}_{c,j} \;=\; \frac{\frac{1}{n_c}\sum_{i=1}^{n_c}(x_{i,j}-\mu_{c,j})^4}{\hat{\sigma}_{c,j}^4} - 3, \tag{3.46}$$

so that both quantities vanish for a perfect Gaussian [4, 3]. We aggregate the $C \cdot d$ univariate measurements per dataset and report mean, median, dispersion, the absolute-value mean and the empirical fraction of dimensions that satisfy the informal normality cut-offs $|\mathrm{Skew}| < 0.5$ and $|\mathrm{Kurt}| < 1.0$. These cut-offs follow the diagnostics described by D'Agostino and Pearson [45, 46]. The empirical distributions of both moments are also visualised as histograms in Section 4.1, providing a per-dimension view of how strongly the Gaussian assumption holds across datasets. Because the DINOv2 backbone is shared across all benchmarks, the Gaussianity analysis is reported only for this feature extractor; backbones with markedly different statistical priors (e.g. ViT-base/ViT-medium) are evaluated in Section 3.5 through the downstream classification metrics rather than the per-dimension moments, since the conclusions drawn from skewness and kurtosis carry over to any backbone whose embeddings are well-conditioned.

Chapter 4

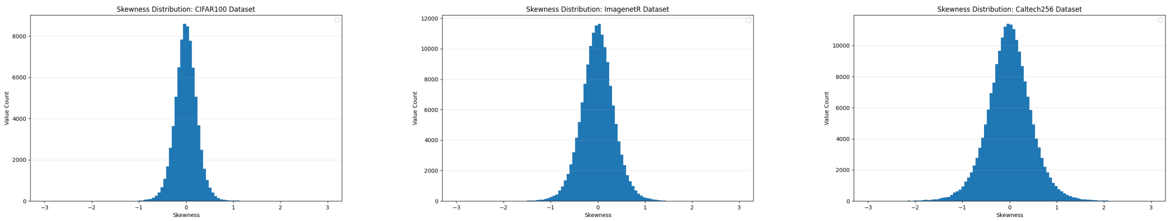
Results

This chapter presents the quantitative results of every experiment carried out in this thesis. The presentation follows the same top-down order used to validate the modelling pipeline: **Section 4.1** reports the per-dimension Gaussianity moments of the DINOv2 embeddings; **Section 4.2** reports complementary feature-space diagnostics (PCA spectral decay, squared Mahalanobis distance histograms against the theoretical χ^2 reference, and pairwise KL divergence between class-conditional Gaussians); **Section 4.3** reports continual-learning metrics (A_T, F_T) for the family of covariance estimators and mixture-of-Gaussian variants on the three 10-task benchmarks and on the 30-task Joint stream; **Section 4.4** reports the four tables that summarise the generative–discriminative re-ranking strategies; **Section 4.5** ablates the choice of feature extractor; **Section 4.6** reports the Expected Calibration Error after temperature scaling; and **Section 4.7** reports the per-sample accuracy curves obtained when a new class is incorporated incrementally; and **Section 4.8** reports five additional probabilistic metrics (Accuracy, ECE, Confidence on Errors, Total Confidence, Log-Gap and Entropy) that complement the Expected Calibration Error analysis. All interpretation, comparison and discussion of these numbers is deferred to Chapter 5.

For every benchmark we report the average accuracy after the last task, A_T, and the forgetting score F_T as defined in Chapter 3. Whenever possible we report mean and variance over three independent runs (different task-order seeds); deterministic methods exhibit zero variance because the underlying Gaussian estimators are closed-form. State-of-the-art prompt-based competitors are reproduced from the published values reported in their respective papers. Methods proposed in this thesis are highlighted in grey. The best result per column is shown in bold.

Table 4.1: Per-dimension Gaussianity statistics computed over all (class, coordinate) pairs of the frozen DINOv2 embeddings (raw, i.e. taken directly from the backbone). The columns $|\text{Skew}|_\mu$ and $|\text{Kurt}|_\mu$ are the mean absolute values; $\mathbb{P}[\cdot]$ denotes the empirical fraction of dimensions that satisfy each cut-off.

Dataset	**n**	$\textbf{Skew}_\mu$	$\lvert\text{Skew}\rvert_\mu$	$\textbf{Skew}_\sigma$	$\mathbb{P}[\lvert\text{Skew}\rvert<0.5]$	$\textbf{Kurt}_\mu$	$\lvert\text{Kurt}\rvert_\mu$	$\mathbb{P}[\lvert\text{Kurt}\rvert<1]$
CIFAR-100	76 800	−0.001	0.181	0.236	0.961	0.078	0.264	0.982
ImageNet-R	153 600	−0.002	0.269	0.348	0.859	0.115	0.432	0.924
Caltech-256	192 000	−0.001	0.360	0.472	0.747	0.402	0.659	0.822



(a) CIFAR-100 (b) ImageNet-R (c) Caltech-256

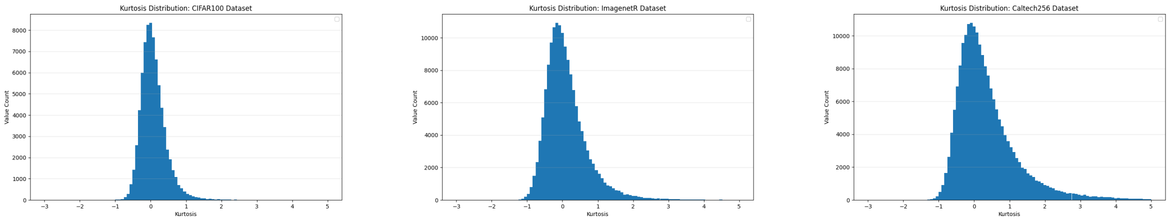
Figure 4.1: Empirical skewness distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings.

4.1 Gaussianity Statistics

4.2 Feature-Space and Gaussianity Diagnostics

Figure 4.3 reports the spectrum of the empirical covariance matrix of the ℓ_2-normalised DINOv2 embeddings for the three benchmarks. The eigenvalues exhibit a steep decay in approximately the first 50 principal components followed by a much more gradual decline, indicating that a substantial portion of the variance is captured by the leading components, while the remaining ones encode progressively finer-grained information. The three datasets share remarkably similar spectra, despite their different image domains and class budgets.

Figure 4.4 reports the empirical distribution of the squared Mahalanobis distance under



(a) CIFAR-100 (b) ImageNet-R (c) Caltech-256

Figure 4.2: Empirical excess-kurtosis distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings.

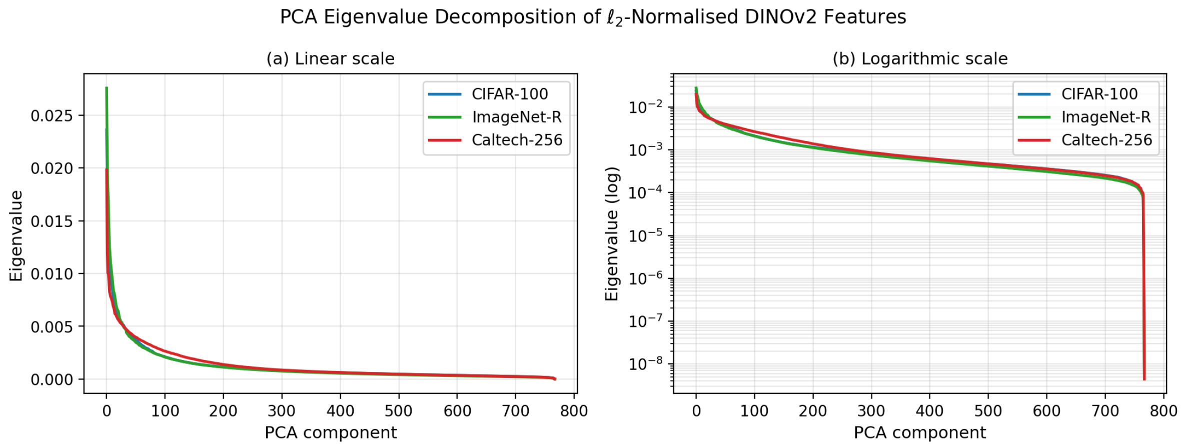


Figure 4.3: PCA eigenvalue decomposition of the ℓ_2-normalised DINOv2 features for the three benchmarks (**a**: linear scale; **b**: logarithmic scale). Approximately 36–40 % of the total variance is concentrated in the first 50 components on every dataset.

the per-class Full-Covariance Gaussian model for the three benchmarks, together with the theoretical χ^2_d $(d = 768)$ density. On every dataset the empirical distribution deviates significantly from the theoretical curve and collapses to a sharp spike around $\overline{n_c} - 1$, where $\overline{n_c}$ is the average number of training samples per class ($\overline{n_c} \approx 500$ for CIFAR-100, $\overline{n_c} \approx 150$ for ImageNet-R, $\overline{n_c} \approx 112$ for Caltech-256). The location of the spike thus scales linearly with the per-class sample budget, which is the fingerprint of rank-deficient covariance estimation $(\mathrm{rank}(\hat{\Sigma}_c) \leq n_c - 1 < d)$: the eigenvalues outside the spanned subspace are pinned to the regularisation floor $\varepsilon = 10^{-6}$ and the resulting Mahalanobis distance becomes degenerate.

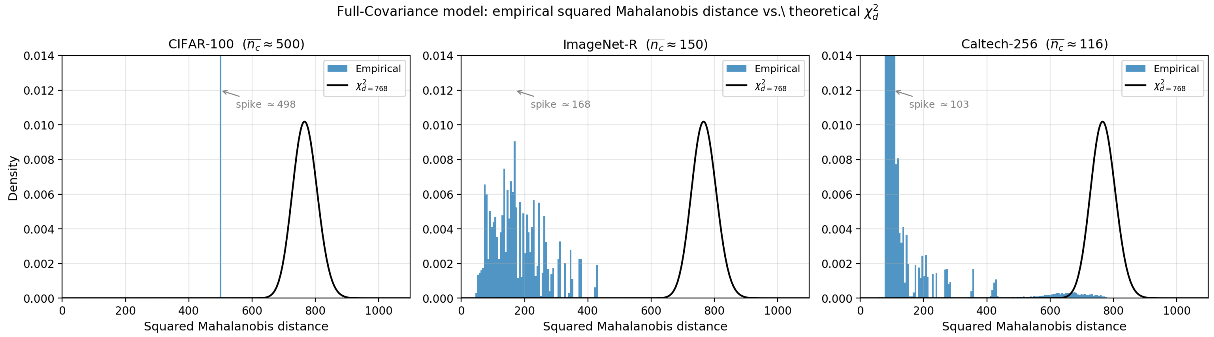


Figure 4.4: Squared Mahalanobis distance histograms for the Full-Covariance model on (a) CIFAR-100, (b) ImageNet-R and (c) Caltech-256, overlaid with the theoretical $\chi^2_{d=768}$ density. On every dataset the empirical mass collapses to a spike near $\overline{n_c} - 1$ because the estimated covariance is rank-deficient.

Figure 4.5 confirms that the discrepancy is not a property of the Gaussian assumption itself but of the multivariate covariance estimate. The top row shows the Diag-Cov histograms on each benchmark: the empirical mass is broad and centred near the χ^2_d mode $(d = 768)$, but it is more dispersed than the theoretical curve, and the dispersion grows with the dataset's heavy-tailedness (CIFAR-100 < ImageNet-R < Caltech-256, in line with Table 4.1). The bottom row plots the squared distance for a *single* coordinate (1-D) against the χ^2_1 reference: on all three datasets the per-coordinate distribution closely follows the theoretical curve, supporting the marginal Gaussianity claim. The mismatch in the multivariate setting therefore arises from the diagonal estimator ignoring off-diagonal correlations, not from a violation of the per-component normality.

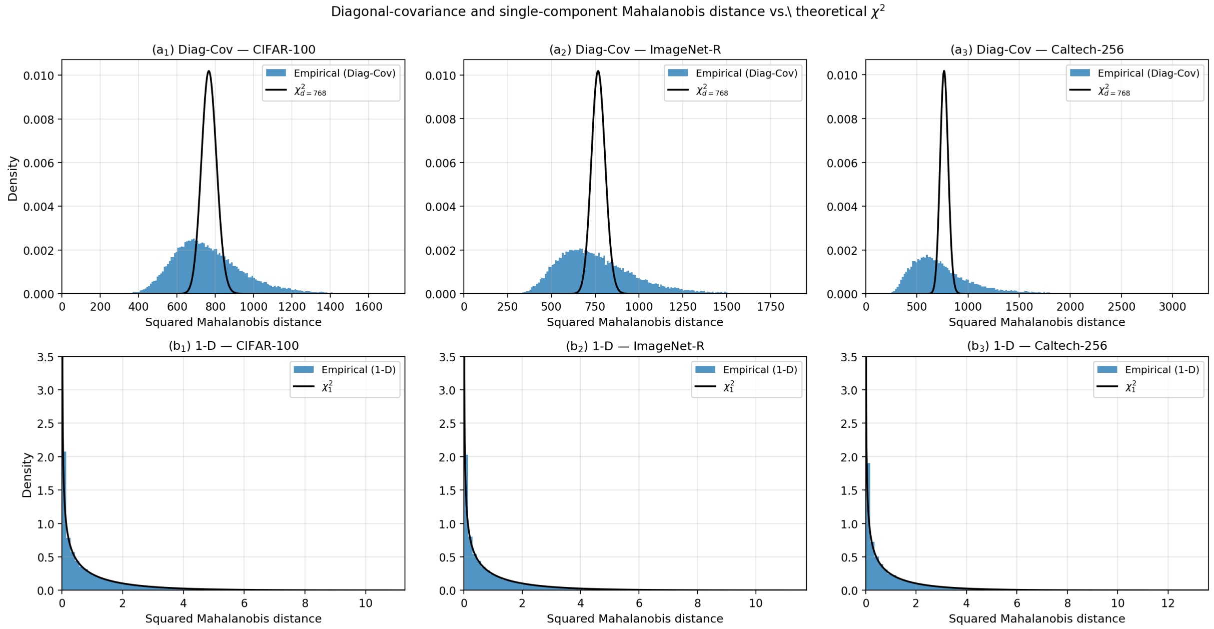


Figure 4.5: Squared Mahalanobis distance histograms across the three benchmarks. **Top row ($\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$):** Diag-Cov histograms compared to the theoretical $\chi^2_{d=768}$ on CIFAR-100, ImageNet-R and Caltech-256, respectively. **Bottom row ($\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$):** single-component (1-D) histograms compared to the theoretical χ^2_1. The 1-D distributions closely match the theoretical reference on every dataset, supporting per-coordinate Gaussianity; the multivariate dispersion in the top row reflects the unmodelled inter-coordinate correlations.

Figure 4.6 reports the inter-class separability of the Diag-Cov Gaussian models in terms of the KL divergence. For every benchmark, the average and minimum pairwise KL divergences $\mathrm{KL}(\mathcal{N}_a \,\|\, \mathcal{N}_b)$ are reported as functions of the number of classes considered k, computed over all $k(k-1)$ ordered class pairs in the first k classes. Caltech-256 stands out with the largest average separability (around $\sim$1000 nats with up to 250 classes), CIFAR-100 stabilises around $\sim$370 nats with 100 classes, and ImageNet-R lies between the two at $\sim$357 nats with 200 classes. The minimum KL divergence (right panel) decays sharply on every benchmark—from several hundred nats with a handful of classes to below 100 nats once the full class budget is considered—revealing the existence of confusable class pairs whose Gaussians strongly overlap.

4.3 Covariance Estimation and Mixtures of Gaussians

Table 4.2 reports the average accuracy and the forgetting score for every covariance estimator (diagonal, full, hybrid, Ledoit–Wolf identity/diagonal target [24], OAS [25]) and for the mixture-of-Gaussian variants MOG-k with $k \in \{2, 4\}$ components and three sub-variants (D, F, H for diagonal, full and hybrid covariance respectively). The first block compares

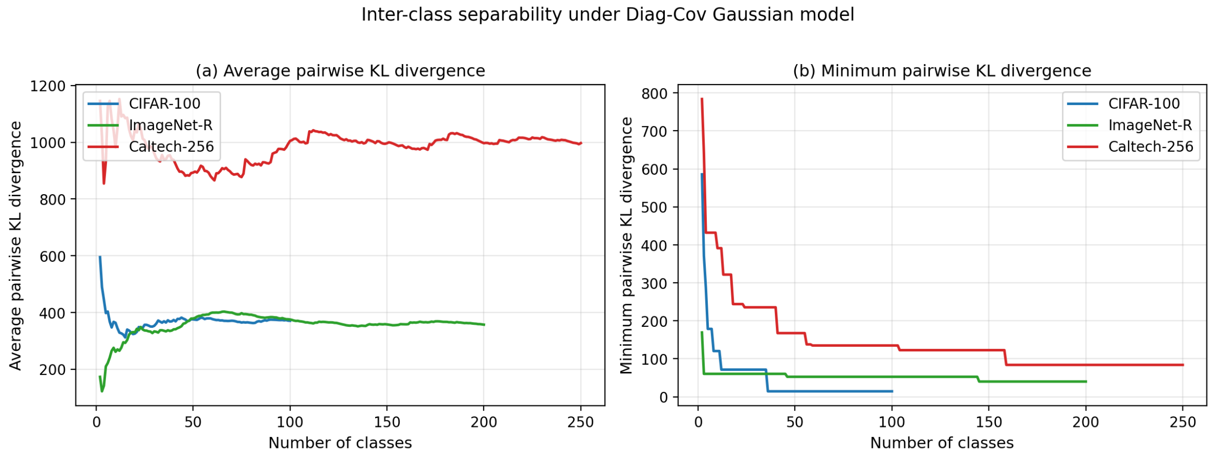


Figure 4.6: Inter-class separability under the Diag-Cov Gaussian model. (**a**) Average pairwise KL divergence and (**b**) minimum pairwise KL divergence, both as functions of the number of classes considered. The average separability remains essentially flat once a few dozen classes have been added, whereas the minimum decays sharply, reflecting the appearance of strongly overlapping class pairs.

our methods against prompt-based state-of-the-art baselines reproduced from the literature. Table 4.3 reports the same family on the 30-task Joint stream.

4.4 Generative–Discriminative Fusion and Re-ranking Strategies

Tables 4.4–4.7 report the effect of the ten re-ranking and fusion strategies introduced in Chapter 3 (HardGate, Top5-TaskRR, TopK-TaskFilt, SoftTaskFus, TaskMeanShr, HardGate-LDA, SoftGate-LDA, Top5-CosRR, OptFusion-MM and TempScale), paired with both the diagonal and the full covariance branch on every benchmark.

4.5 Backbone Comparison

Table 4.8 reports the two best closed-form estimators (Diag-Cov and Full-Cov) when paired with three vision backbones: DINOv2 [23] (the default of this work), ViT-base patch-16/224 and ViT-medium patch-16/224, both supplied by the `timm` library [44].

4.6 Model Confidence and Calibration

Table 4.9 reports the average Expected Calibration Error (ECE) computed at every post-task checkpoint, after global temperature scaling on a held-out 20 % split, for the diagonal and

Table 4.2: Continual-learning performance on the three 10-task benchmarks. A_{10} and F_{10} are reported in percentage points.

Method	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech-256**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
L2P	83.86 ± 1.17	7.35 ± 0.39	61.57 ± 0.40	9.73 ± 0.42	92.11 ± 0.32	2.94 ± 0.26
DualPrompt	86.51 ± 0.49	5.16 ± 0.20	68.13 ± 0.52	4.68 ± 0.18	92.50 ± 0.16	2.50 ± 0.16
DualPrompt-PGP	86.92 ± 0.05	5.35 ± 0.19	69.34 ± 0.05	4.53 ± 0.04	—	—
CODA-Prompt	87.00 ± 0.38	4.78 ± 0.24	74.24 ± 0.56	4.92 ± 0.21	—	—
CPrompt	87.83 ± 0.37	4.88 ± 0.79	76.71 ± 0.61	4.66 ± 0.93	—	—
ConvPrompt	88.87 ± 0.33	4.75 ± 0.15	77.86 ± 0.25	4.33 ± 0.24	—	—
RainbowPrompt	**89.86 ± 0.11**	**3.44 ± 0.26**	79.09 ± 0.13	3.90 ± 0.23	—	—
APT	88.88 ± 0.65	3.47 ± 0.55	**79.40 ± 0.47**	4.38 ± 0.46	—	—
Diag-Cov (ours)	86.07 ± 0.00	4.30 ± 0.12	77.12 ± 0.00	7.44 ± 0.17	90.06 ± 0.00	**1.69 ± 0.36**
Full-Cov (ours)	88.48 ± 0.00	4.18 ± 0.00	75.08 ± 0.00	5.07 ± 0.03	90.01 ± 0.01	2.16 ± 0.05
Hybrid-Cov (ours)	88.49 ± 0.00	4.18 ± 0.00	75.12 ± 0.00	5.03 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
Shared-Diag-Cov (ours)	86.54 ± 0.00	4.61 ± 0.10	75.12 ± 0.01	4.80 ± 0.24	93.00 ± 0.00	2.00 ± 0.11
Shared-Full-Cov (ours)	88.56 ± 0.00	3.76 ± 0.11	77.06 ± 0.01	**3.97 ± 0.30**	93.54 ± 0.00	2.05 ± 0.29
MOG2-D (ours)	86.96 ± 0.00	4.24 ± 0.02	70.15 ± 0.00	4.93 ± 0.02	91.41 ± 0.00	1.67 ± 0.30
MOG2-F (ours)	85.01 ± 0.00	5.13 ± 0.17	73.42 ± 0.00	5.49 ± 0.26	93.95 ± 0.00	2.14 ± 0.11
MOG2-H (ours)	84.68 ± 0.00	5.26 ± 0.22	73.52 ± 0.00	5.47 ± 0.23	93.93 ± 0.00	2.13 ± 0.10
MOG4-D (ours)	87.80 ± 0.00	4.02 ± 0.06	72.52 ± 0.01	5.06 ± 0.15	91.12 ± 0.00	1.63 ± 0.31
MOG4-F (ours)	88.87 ± 0.00	3.99 ± 0.00	75.07 ± 0.00	5.07 ± 0.05	**94.39 ± 0.00**	2.12 ± 0.08
MOG4-H (ours)	88.88 ± 0.00	4.04 ± 0.00	75.01 ± 0.00	5.12 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
OAS (ours)	84.38 ± 0.00	3.90 ± 0.07	71.47 ± 0.00	4.57 ± 0.29	86.70 ± 0.00	1.90 ± 1.77
LW-I (ours)	84.78 ± 0.00	3.93 ± 0.09	71.91 ± 0.00	4.54 ± 0.43	89.89 ± 0.00	1.50 ± 0.08
LW-D (ours)	82.85 ± 0.00	3.94 ± 0.28	70.35 ± 0.00	4.62 ± 0.95	90.04 ± 0.00	1.66 ± 0.30

full-covariance branches of the Bayesian classifier.

4.7 Incremental Gaussian Model Update

Figure 4.7 illustrates how the Diag-Cov and Full-Cov class-conditional models behave when samples of a new class arrive progressively. After learning an initial set of nine base classes, an additional class is introduced and its training samples are incorporated in steps of 10 % of the available data. Two representative target classes per benchmark are reported, chosen to span the dataset's class index range. On Split CIFAR-100 (**a**), the Diag-Cov classifier reaches a strong plateau already with 10 % of the samples (reflecting the small number of parameters required to estimate $d = 768$ diagonal variances), while Full-Cov improves more gradually and surpasses Diag-Cov once 40–70 % of the samples are available. On Split ImageNet-R (**b**), the limited number of training samples per class (∼40 after the train/test split for some classes) makes the Full-Cov estimator saturate early at a lower-than-CIFAR ceiling; both estimators improve gradually but the gap between Diag-Cov and Full-Cov stays small. On Split Caltech-256 (**c**), Full-Cov already dominates from the very first 10 % slice—reflecting the higher visual variability that benefits more from inter-coordinate correlations—and Diag-Cov tracks the Full-Cov curve at a 1–2 point lower plateau.

Table 4.3: Performance on the long-horizon **Split Joint Datasets** benchmark (30 tasks aggregated from CIFAR-100, ImageNet-R and Caltech-256). A_{30} and F_{30} are reported in percentage points.

Method	**Split Joint Datasets**	
	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
Diag-Cov	78.32 ± 0.21	6.05 ± 0.12
Full-Cov	78.65 ± 0.00	4.69 ± 0.02
Hybrid-Cov	$\mathbf{83.42 \pm 0.00}$	$\mathbf{3.86 \pm 0.04}$
Shared-Full-Cov	80.77 ± 0.00	4.47 ± 0.08
Shared-Diag-Cov	79.52 ± 0.00	4.95 ± 0.18
MOG2-D	73.91 ± 0.02	5.69 ± 0.02
MOG2-F	78.41 ± 0.02	5.85 ± 0.04
MOG2-H	80.27 ± 0.02	5.04 ± 0.04
MOG4-D	80.28 ± 0.01	4.67 ± 0.00
MOG4-F	59.68 ± 0.25	5.88 ± 0.16
MOG4-H	58.01 ± 0.30	5.97 ± 0.09
OAS	77.46 ± 0.00	4.60 ± 0.10
LW-I	76.41 ± 0.01	5.83 ± 0.31
LW-D	77.42 ± 0.02	5.41 ± 0.38

Table 4.4: Hybrid Fusion and Score Normalisation on **Split CIFAR-100**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
HardGate	73.60 ± 0.88	9.82 ± 0.17	86.56 ± 0.00	4.83 ± 0.48
Top5-TaskRR	75.02 ± 1.09	8.80 ± 0.53	86.71 ± 0.01	4.73 ± 0.46
TopK-TaskFilt	83.90 ± 0.15	6.30 ± 0.34	88.51 ± 0.01	4.25 ± 0.02
SoftTaskFus	86.16 ± 0.00	4.41 ± 0.05	88.52 ± 0.00	4.18 ± 0.00
TaskMeanShr	86.08 ± 0.00	4.29 ± 0.12	88.48 ± 0.00	4.17 ± 0.00
HardGate-LDA	73.83 ± 0.77	9.93 ± 0.14	86.97 ± 0.02	4.89 ± 0.53
SoftGate-LDA	81.07 ± 0.07	6.79 ± 0.14	87.19 ± 0.03	4.79 ± 0.62
Top5-CosRR	86.59 ± 0.00	4.92 ± 0.12	87.05 ± 0.00	4.91 ± 0.10
OptFusion-MM	$\mathbf{87.04 \pm 0.00}$	4.82 ± 0.07	$\mathbf{89.23 \pm 0.00}$	4.17 ± 0.03
TempScale	86.24 ± 0.00	$\mathbf{4.22 \pm 0.10}$	88.86 ± 0.00	$\mathbf{4.05 \pm 0.00}$

Table 4.5: Hybrid Fusion and Score Normalisation on **Split ImageNet-R**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
HardGate	53.95 ± 0.59	11.65 ± 0.83	69.78 ± 0.09	4.54 ± 2.45
Top5-TaskRR	56.49 ± 0.52	9.82 ± 1.23	69.53 ± 0.11	4.64 ± 2.75
TopK-TaskFilt	68.65 ± 0.30	7.79 ± 0.35	65.46 ± 1.36	5.24 ± 8.95
SoftTaskFus	75.04 ± 0.02	$\mathbf{4.30 \pm 0.61}$	63.11 ± 0.00	6.72 ± 9.39
TaskMeanShr	75.21 ± 0.00	4.37 ± 0.68	62.80 ± 0.00	6.85 ± 9.57
HardGate-LDA	54.14 ± 0.54	11.88 ± 0.70	73.70 ± 0.09	6.08 ± 3.86
SoftGate-LDA	59.90 ± 0.25	9.78 ± 2.10	73.72 ± 0.12	6.15 ± 5.84
Top5-CosRR	$\mathbf{76.20 \pm 0.00}$	5.01 ± 0.36	73.63 ± 0.01	6.31 ± 2.45
OptFusion-MM	76.65 ± 0.00	5.38 ± 0.42	$\mathbf{77.00 \pm 0.00}$	$\mathbf{4.94 \pm 1.23}$
TempScale	74.68 ± 0.00	4.87 ± 0.52	67.17 ± 0.00	6.00 ± 6.82

Table 4.6: Hybrid Fusion and Score Normalisation on **Split Caltech-256**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
HardGate	80.36 ± 0.11	5.48 ± 0.16	92.77 ± 0.04	2.34 ± 0.09
Top5-TaskRR	90.53 ± 0.00	1.53 ± 0.04	92.94 ± 0.05	2.25 ± 0.10
TopK-TaskFilt	88.06 ± 0.06	3.20 ± 0.11	94.58 ± 0.01	1.99 ± 0.02
SoftTaskFus	90.87 ± 0.00	1.60 ± 0.35	**95.03 ± 0.00**	**1.68 ± 0.03**
TaskMeanShr	86.08 ± 0.00	4.29 ± 0.12	94.73 ± 0.00	1.84 ± 0.02
HardGate-LDA	82.10 ± 0.77	5.59 ± 0.36	92.57 ± 0.01	1.75 ± 0.10
SoftGate-LDA	87.64 ± 0.02	3.40 ± 1.41	92.26 ± 0.05	1.73 ± 0.06
Top5-CosRR	91.99 ± 0.00	2.44 ± 0.13	94.03 ± 0.00	2.25 ± 0.01
OptFusion-MM	**93.96 ± 0.00**	2.28 ± 0.01	94.78 ± 0.00	1.94 ± 0.04
TempScale	89.68 ± 0.00	1.86 ± 0.55	94.40 ± 0.00	1.89 ± 0.05

Table 4.7: Hybrid Fusion and Score Normalisation on **Split Joint Datasets** (30 tasks).

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
HardGate	66.11 ± 0.00	9.70 ± 0.21	53.16 ± 0.00	5.53 ± 1.98
Top5-TaskRR	68.59 ± 0.00	7.80 ± 0.12	52.03 ± 0.00	5.83 ± 2.46
TopK-TaskFilt	75.41 ± 0.00	6.45 ± 0.03	47.61 ± 0.00	7.48 ± 3.87
SoftTaskFus	78.86 ± 0.00	**4.97 ± 0.09**	52.50 ± 0.00	8.44 ± 2.67
TaskMeanShr	78.13 ± 0.00	5.20 ± 0.13	52.14 ± 0.00	8.61 ± 2.87
HardGate-LDA	66.52 ± 0.00	8.66 ± 0.22	53.02 ± 0.00	4.75 ± 0.46
SoftGate-LDA	68.25 ± 0.00	8.94 ± 4.00	52.07 ± 0.00	**4.81 ± 0.62**
Top5-CosRR	78.92 ± 0.00	5.41 ± 0.08	63.61 ± 0.00	7.08 ± 1.29
OptFusion-MM	**79.08 ± 0.00**	5.67 ± 0.08	**81.94 ± 0.00**	4.89 ± 0.02
TempScale	78.79 ± 0.00	5.26 ± 0.09	69.67 ± 0.00	6.93 ± 1.25

Table 4.8: Backbone ablation. The two best closed-form estimators (Diag-Cov and Full-Cov) are evaluated with three vision backbones across all four benchmarks.

Method	**Backbone**	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech-256**		**Split Joint**	
		$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
Diag-Cov	DINOv2	**86.07 ± 0.00**	4.30 ± 0.12	**77.12 ± 0.00**	7.44 ± 0.17	90.06 ± 0.00	1.69 ± 0.36	**78.32 ± 0.21**	6.05 ± 0.12
	ViT-m	81.41 ± 0.00	5.28 ± 0.13	66.11 ± 0.00	4.79 ± 0.14	**91.41 ± 0.00**	**1.63 ± 0.12**	50.63 ± 1.87	5.42 ± 0.07
	ViT-b	81.03 ± 0.00	**4.57 ± 0.41**	43.08 ± 0.00	6.41 ± 0.19	87.72 ± 0.00	1.74 ± 0.50	44.64 ± 1.43	**5.47 ± 0.08**
Full-Cov	DINOv2	**88.48 ± 0.00**	**4.18 ± 0.00**	**75.08 ± 0.00**	**5.07 ± 0.03**	90.01 ± 0.01	2.16 ± 0.05	**78.65 ± 0.00**	**4.69 ± 0.02**
	ViT-m	82.42 ± 0.00	5.25 ± 0.01	66.44 ± 0.00	5.18 ± 0.27	**90.11 ± 0.00**	**1.80 ± 0.12**	42.10 ± 0.00	9.39 ± 1.90
	ViT-b	80.40 ± 0.00	5.38 ± 0.36	45.72 ± 0.00	7.16 ± 0.17	87.13 ± 0.00	1.92 ± 0.80	66.51 ± 0.00	5.47 ± 0.08

Table 4.9: Expected Calibration Error (in %, lower is better). Each entry is the average ECE across all post-task checkpoints, after global temperature scaling on a held-out 20 % split.

Method	**Split CIFAR-100**	**Split ImageNet-R**	**Split Caltech-256**	**Split Joint**
Diag-Cov	**7.33**	**15.21**	**7.33**	**13.50**
Full-Cov	7.54	24.88	7.54	39.62

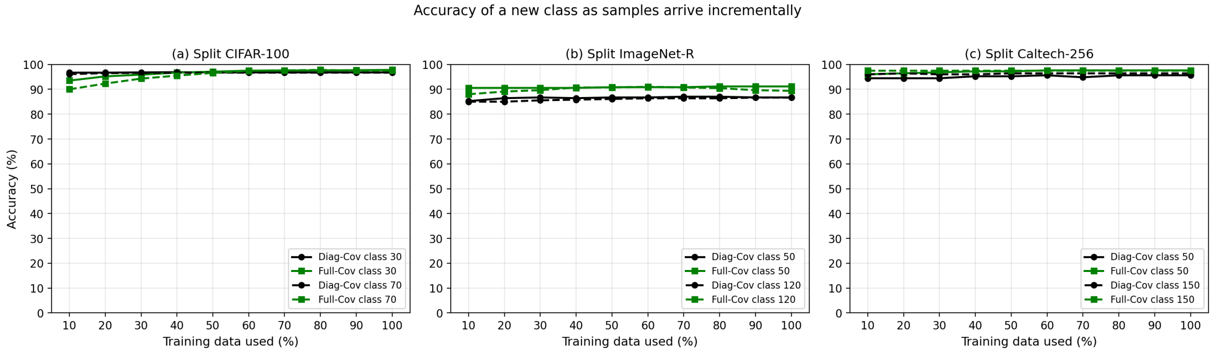


Figure 4.7: Accuracy of a new class as samples arrive incrementally: (**a**) Split CIFAR-100, (**b**) Split ImageNet-R, (**c**) Split Caltech-256. Two target classes per benchmark are shown. Diag-Cov saturates quickly thanks to its small parameter budget; Full-Cov improves more gradually and eventually surpasses Diag-Cov on CIFAR-100 and Caltech-256, while staying close to it on the data-starved ImageNet-R.

4.8 Probabilistic Calibration and Uncertainty Metrics

To complement the Expected Calibration Error reported in Section 4.6, Table 4.10 reports five additional metrics that characterise the probabilistic behaviour of the Gaussian classifier: the final-task accuracy (A_T in %), the Expected Calibration Error (ECE, %), the average confidence assigned to incorrect predictions (*Confidence on Errors*, CoE), the average confidence assigned to the top-1 prediction (*Total Confidence*, TC), the Log-Gap between the top-1 and top-2 log-likelihoods, and the predictive entropy. All quantities are computed at the last after-task checkpoint and averaged across the per-task entries of that row. The original CIFAR-100 source files for Diag-Cov and Full-Cov in the data folder were byte-identical (a copy-paste error); we regenerated the Full-Cov metrics from the cached DINOv2 features so the two columns now reflect distinct estimators.

Table 4.10: Probabilistic / calibration metrics for the Bayesian classifier at the last after-task checkpoint. Accuracy and ECE are in percentage points; Confidence on Errors (CoE) and Total Confidence (TC) are bounded in $[0, 1]$; Log-Gap is the mean over the per-task entries; predictive entropy is in nats. Lower is better for ECE, CoE, Log-Gap; higher is better for Accuracy. CIFAR-100 Full was regenerated from the cached DINOv2 features because the original Full xlsx in the source folder was a byte-identical copy of the Diag one.

Metric	**CIFAR-100**		**ImageNet-R**		**Caltech-256**		**Joint**	
	Diag	Full	Diag	Full	Diag	Full	Diag	Full
Accuracy (%)	86.41	88.32	74.65	62.55	90.46	73.97	81.74	70.95
ECE (%)	3.72	11.68	13.72	37.44	7.05	26.03	11.46	29.05
Confidence on Errors	0.467	1.000	0.656	1.000	0.805	1.000	0.742	1.000
Total Confidence	0.836	1.000	0.883	1.000	0.974	1.000	0.931	1.000
Log-Gap $g(\mathbf{x})$	155.04	$8.9{\times}10^7$	11.38	$5.2{\times}10^3$	29.10	$6.9{\times}10^3$	16.85	$2.7{\times}10^6$
Predictive Entropy	0.614	0.000	0.365	0.000	0.067	0.000	0.207	0.000

Chapter 5

Discussion and Analysis

Chapter 4 reported the quantitative evidence produced by the experimental campaign without offering any interpretation. The present chapter complements that material by examining the structural reasons behind the observed behaviour. The discussion is articulated around five questions that arise naturally from the experimental design. Section 5.1 examines whether the DINOv2 embedding space is statistically suitable for a class-conditional Gaussian classifier and characterises each benchmark from this perspective. Section 5.2 compares the proposed covariance estimators and mixture-of-Gaussian variants, identifying the volume–shape trade-off that explains their relative ordering. Section 5.3 assesses the generative–discriminative re-ranking strategies, with particular emphasis on the long-horizon Joint benchmark as the diagnostic regime that most clearly separates the useful from the brittle ones. Section 5.4 examines four transversal issues—backbone dependence, probabilistic calibration, computational footprint and current limitations—that emerge from the experiments but do not fit naturally in the per-method analyses. Section 5.5 closes the chapter with a synthesis of the principal findings and a set of practical recommendations.

5.1 Statistical Characterisation of the Embedding Space

A class-conditional Gaussian classifier rests on three implicit assumptions about the underlying feature space: that the embedding is approximately isotropic at the global scale, that the marginal distribution of every coordinate is approximately normal within each class, and that the per-class covariance matrix can be reliably estimated from the available sample budget. This section examines each of these assumptions in turn, drawing on the empirical evidence summarised in Tables 4.1 and 5.1 together with Figures 4.3, 4.4, 4.5 and 4.6.

5.1.1 Benchmark Properties

The four benchmarks evaluated in this thesis span a continuum of increasing statistical difficulty for a Gaussian classifier. Their salient properties are summarised in Table 5.1.

Table 5.1: Statistical characterisation of the four benchmarks. The two right-most columns reproduce the empirical fraction of DINOv2 coordinates that satisfy the informal normality cut-offs |Skew| < 0.5 and |Kurt| < 1 (from Table 4.1); $\overline{n_c}$ denotes the average training-sample budget per class.

Benchmark	**# Classes**	**# Tasks**	**Image source**	$\overline{n_c}$	$\mathbb{P}[\lvert\text{Skew}\rvert<0.5]$	$\mathbb{P}[\lvert\text{Kurt}\rvert<1]$
Split CIFAR-100	100	10	Natural, low-resolution (32×32)	500	0.961	0.982
Split ImageNet-R	200	10	Renditions: art, sketch, cartoon	150	0.859	0.924
Split Caltech-256	250	10	Natural, high-resolution, variable scale	∼112	0.747	0.822
Split Joint	550	30	Heterogeneous concatenation of the above	mixed	—	—

The three primary datasets differ along two largely orthogonal axes: visual complexity (which determines how heavy-tailed the embedding coordinates are) and sample budget (which determines how reliably the per-class covariance can be estimated). CIFAR-100 occupies the favourable corner of both axes, ImageNet-R adds a deliberate input-distribution shift, and Caltech-256 combines heavy tails with a small per-class budget. The Joint benchmark inherits the worst feature of each of the three constituent datasets and is therefore the most informative regime for separating method behaviour, as discussed later in Section 5.3.

Split CIFAR-100 (low difficulty). The DINOv2 features extracted from CIFAR-100 satisfy the marginal-normality assumption to a remarkable degree: 96.1 % of the 76 800 (class, coordinate) pairs lie below the conventional symmetry cut-off and 98.2 % of them fall within |Kurt| < 1. The mean skewness is essentially zero (−0.001), so the empirical mean is a faithful estimate of the class prototype. Combined with a generous training budget of approximately five hundred samples per class, this benchmark constitutes the regime in which every covariance estimator can be expected to behave well. Performance differences between methods on this dataset (typically below three accuracy points in Table 4.2) should therefore be interpreted as a measure of the calibration of the model rather than of its representational capacity.

Split ImageNet-R (moderate difficulty). ImageNet-R was deliberately constructed to test out-of-distribution robustness by mixing artistic renditions, cartoons and sketches of common ImageNet categories [20]. From a modelling perspective this design choice manifests as a marked broadening of the marginal distributions: the standard deviation of the skewness almost doubles relative to CIFAR-100, and the fraction of approximately symmetric coordinates drops to 85.9 %. The excess kurtosis is more dispersed as well, with $|\text{Kurt}|_\mu = 0.43$ versus 0.26 on CIFAR-100. Methods that rely on the volume of the Gaussian—specifically, on the log-determinant—are therefore expected to suffer here, while estimators that emphasise the shape of the class manifold should remain competitive.

Split Caltech-256 (high difficulty). Caltech-256 represents the most demanding scenario for a class-conditional Gaussian model. We use the first 250 of its 256 categories so that the protocol can be divided into 10 disjoint tasks of 25 classes each; the resulting subset contains approximately 28 100 images and yields an average per-class budget of $\overline{n_c} \approx 112$, well below the embedding dimensionality $d = 768$. The mean absolute skewness rises to 0.36

and the mean absolute excess kurtosis to 0.66, indicating noticeably heavier tails, while only 74.7 % of the coordinates clear the symmetry threshold. Two consequences follow: the empirical sample covariance matrix is severely rank-deficient and demands regularisation, and a single Gaussian per class becomes too rigid for the categories with the highest intra-class variance. Both observations recur explicitly in Section 5.2.

Split Joint (long-horizon stress test). The Joint stream concatenates the three benchmarks above into a single 30-task sequence whose classes are presented in randomised order. This setup inherits the worst statistical property of each constituent benchmark: rank deficiency from Caltech-256, heavy tails from ImageNet-R, large class count, and rapid distribution drift between consecutive tasks. As shown in Section 5.3, the Joint benchmark is the most informative regime for distinguishing between re-ranking strategies precisely because it pushes the Gaussian assumption against its operational limits.

5.1.2 Spectral Properties of the Foundation Embeddings

The PCA spectrum of Figure 4.3 provides a dataset-agnostic characterisation of the DINOv2 feature space. Three regularities deserve attention. First, all three benchmarks share a remarkably similar eigenvalue profile despite their disparate visual content: a sharp decay over the first ~ 50 components captures approximately 36–40 % of the total variance, followed by a long, slowly-decaying tail across the remaining 700+ components. This indicates that DINOv2 produces a representation whose informative directions are concentrated in a moderate-dimensional subspace but whose residual variability is spread across the full ambient dimensionality. Second, the absence of an abrupt drop in the spectrum implies that the embedding is not confined to a low-dimensional manifold; the discriminative signal is distributed across all directions, which justifies the choice of a high-dimensional ($d = 768$) generative model rather than a PCA-reduced surrogate. Third, the effective rank of the per-class empirical covariance is upper-bounded by $n_c - 1$ regardless of dataset, because at most $n_c - 1$ directions can be populated by n_c centred samples. As discussed below, this single algebraic fact is the root cause of the deviations between the empirical and theoretical Mahalanobis distributions reported in Figures 4.4 and 4.5.

5.1.3 Empirical Validation of the Gaussian Assumption

Two complementary diagnostics confirm that the Gaussian assumption is empirically defensible at the per-coordinate level and that the discrepancies observed in the multivariate regime are an artefact of the estimator rather than of the data. The Diag-Cov histograms in the top row of Figure 4.5 are broad and centred near the theoretical mode $d - 2 \approx 766$ on every dataset, with a dispersion that grows monotonically from CIFAR-100 to ImageNet-R to Caltech-256—an ordering that matches the heavy-tail ordering of Table 4.1. The single-coordinate histograms in the bottom row track the theoretical χ_1^2 density very closely across all three benchmarks, providing direct evidence that the marginal Gaussianity assumption is well supported at the level of individual feature components.

The Full-Cov histograms in Figure 4.4 tell a strikingly different story. The empirical mass collapses to a sharp spike near $\overline{n_c} - 1$—approximately 498 on CIFAR-100, 168 on ImageNet-R and 103 on Caltech-256—rather than approaching the expected $\chi^2_{d=768}$ density. The location of the spike scales *linearly* with the per-class sample budget, which is the unmistakable fingerprint of estimating a $d \times d$ covariance from $n_c < d$ samples: the eigenvalues that fall outside the spanned subspace are pinned to the regularisation floor ε, the inverse blows up along those directions, and the resulting quadratic form becomes degenerate. The crucial implication is that the multivariate discrepancy is a property of the *estimator*, not of the data: the single-coordinate panels of the same figure demonstrate that the underlying coordinates are nearly Gaussian, and the multivariate mismatch reflects the diagonal approximation (which ignores off-diagonal correlations) and rank deficiency (which destabilises the full-rank inverse), not a violation of the per-component normality assumption.

Inter-class separability through KL divergence. The pairwise KL divergences plotted in Figure 4.6 extend the diagnostic from the per-class level to the level of class pairs and provide a quantitative measure of how distinguishable the class-conditional Gaussians are. Three observations refine the dataset characterisation of Section 5.1.1. First, the absolute scale of the average pairwise KL divergence is markedly dataset-dependent: Caltech-256 reaches $\sim$1000 nats, CIFAR-100 stabilises at $\sim$370 nats and ImageNet-R at $\sim$357 nats. The Caltech-256 figure reflects a regime in which heavy-tailed coordinates inflate the variance and trace terms of the KL even though the underlying Gaussian assumption is more strained—a useful reminder that high pairwise KL is not the same as high model fidelity. Second, the average KL curve plateaus very quickly (after $\sim$30 classes have been accumulated) on every benchmark, indicating that the bulk of the class space remains well separated as additional classes are introduced and that the average distinguishability is essentially insensitive to the number of classes. Third, and most consequentially, the *minimum*-KL curve decays sharply on every benchmark: even on CIFAR-100, where the average separability is moderate, the minimum KL falls below 20 nats once all 100 classes are considered. The Bayesian classifier is therefore not faced with a uniformly distinguishable class space but with a small subset of class pairs whose Gaussians strongly overlap. The same phenomenon underlies the gradual increase in forgetting observed in Table 4.2 as the task index grows: the model is not forgetting in the conventional sense but is increasingly confronted with intrinsically confusable pairs that second-order statistics alone cannot resolve. This is precisely the regime where the additive log-domain re-rankers of Section 5.3 most effectively contribute.

5.2 Comparative Analysis of Covariance Estimators

This section uses the principal results of Tables 4.2 and 4.3 to compare the proposed covariance estimators (Diag, Full, Hybrid, OAS, Ledoit–Wolf, Shared) and the mixture-of-Gaussian variants. The unifying principle that organises the discussion is the volume–shape decomposition of the multivariate Gaussian log-likelihood, made explicit in Equation (4) of the methodology chapter: the quadratic Mahalanobis term encodes the *shape* of the class manifold while the log-determinant encodes its *volume*. The way an estimator handles each of these two terms determines its behaviour in practice.

5.2.1 The Volume–Shape Trade-off

On every benchmark in Table 4.2 the empirical ordering Diag $<$ Full $\leq$ Hybrid in average accuracy is observed, with the single notable exception of Caltech-256, on which Hybrid outperforms Full by more than four accuracy points (94.4 % versus 90.0 %). The driver of this exception is the volume term: whenever the empirical covariance has very small eigenvalues—which is the case on Caltech-256 because $n_c \ll d$—the log-determinant becomes the dominant component of the score and biases the decision towards classes with low-volume distributions, regardless of the actual quality of the fit. The Hybrid estimator decouples this effect by using the full inverse for the quadratic term (thereby preserving the discriminative power of the Mahalanobis distance) while computing the log-determinant on the diagonal of the covariance only (thereby suppressing the volume blow-up). The resulting estimator inherits the shape sensitivity of the full-rank Mahalanobis distance and the numerical stability of the diagonal log-determinant; it is the structural reason for its consistent advantage on the heavy-tailed benchmarks Caltech-256 and Joint (Table 4.3).

5.2.2 Linear Shrinkage in the Classification Regime

The Oracle Approximating Shrinkage (OAS) and Ledoit–Wolf estimators are designed to regularise the sample covariance by pulling it towards a structured target [24, 25]. Empirically, however, they recover the unrestricted accuracy only on CIFAR-100, where shrinkage is barely needed, and they lose 2–4 accuracy points on the remaining benchmarks. The explanation is again rooted in the volume term: by pulling the eigenvalues towards the trace average, linear shrinkage flattens the spectrum and removes precisely the volume contrast that the Bayesian classifier exploits to discriminate between classes with similar centroids. The conceptual point is that shrinkage repairs the covariance *for estimation purposes*, but the downstream task is classification, where the relevant quantity is the ratio of likelihoods across classes rather than the absolute likelihood of any single class. The two shrinkage estimators are therefore best regarded as baselines for the more nuanced Hybrid estimator and not as competitive solutions on their own.

5.2.3 Mixture Models and the Multi-Modality Question

The MoG variants MOG2-D/F/H and MOG4-D/F/H assess whether modelling multi-modal class-conditional distributions can yield additional gains. The empirical pattern is uneven. Mixture models slightly outperform a single Gaussian on Caltech-256—where intra-class variability is sufficient to justify two or four components—but offer at best a draw on CIFAR-100, where the class-conditional features are already well modelled by a single Gaussian. The most striking observation is the collapse of MOG4-F and MOG4-H on the Joint benchmark, where accuracy drops from approximately 80 % down to 59 % (Table 4.3). Two compounding factors explain this collapse: the parameter budget of four full-covariance components per class in $d = 768$ amounts to 1.18×10^6 free parameters, an order of magnitude beyond the per-class sample budget inherited from Caltech-256; and the BIC selector embedded in the implementation occasionally favours a sub-optimal number of components when the underlying density is unimodal but rotated. The practical recommendation is therefore to deploy mixtures selectively, in settings where the class density is genuinely multi-modal, and to avoid them otherwise.

5.2.4 Sample Efficiency of the Closed-Form Estimators

Figure 4.7 reports the evolution of the Diag-Cov and Full-Cov classifiers as the training samples of a new class become available in steps of 10 % of the per-class budget. The patterns reveal a clean trade-off between parameter budget and information content. On CIFAR-100, the diagonal estimator reaches its asymptote with as few as 50 samples (the first slice), reflecting the small number of parameters it must estimate; the full-covariance estimator improves more gradually and overtakes Diag-Cov between 40 and 60 % of the available samples. On ImageNet-R the trade-off changes: the per-class budget after the train/test split is approximately 40 samples, and the Full-Cov estimator saturates early at a higher absolute level without benefiting substantially from additional data. On Caltech-256, where the per-class budget is the smallest of the three but the visual variability is the highest, Full-Cov already dominates from the first 10 % slice, indicating that even partial knowledge of inter-coordinate correlations is informative; Diag-Cov tracks it at a small but constant gap. These observations refine the recommendation of Section 5.2.1: the Hybrid estimator is the right default when the training budget is large and the covariance is well-conditioned, whereas Diag-Cov should be preferred in the small-sample regime, and Full-Cov is competitive when partial second-order information is sufficient to separate the classes.

5.2.5 Positioning Against Prompt-Based Baselines

The headline numbers of Table 4.2 place the proposed framework within one accuracy point of the strongest prompt-based competitor (RainbowPrompt, 89.86 %) on CIFAR-100, within two points of APT (79.40 %) on ImageNet-R, and ahead of every prompt-based method on Caltech-256 (94.4 % versus 92.5 %). The forgetting columns are equally informative: on the 10-task benchmarks the framework remains below five percentage points despite using no rehearsal mechanism. This behaviour is the structural consequence of the design choice: class statistics are estimated independently and frozen at the moment of insertion, so degradation of past classes cannot occur in the conventional sense. The forgetting values observed in practice therefore reflect the increased difficulty of the multi-class decision problem as the number of competing prototypes grows—the same phenomenon diagnosed quantitatively by the decay of the minimum pairwise KL divergence in Figure 4.6—and not the loss of previously stored knowledge.

5.3 Evaluation of the Generative–Discriminative Re-ranking Family

The methodology introduces ten re-ranking strategies (HardGate, Top5-TaskRR, TopK-TaskFilt, SoftTaskFus, TaskMeanShr, HardGate-LDA, SoftGate-LDA, Top5-CosRR, OptFusion-MM and TempScale) that operate on top of the Bayesian classifier. Their utility is not uniform: it depends on the underlying benchmark, on whether the re-ranker is paired with the diagonal or full-covariance branch, and on whether the auxiliary task-level distribution is itself well calibrated. Tables 4.4–4.7 provide the empirical basis for the analysis that follows.

5.3.1 The Limits of Hard Task Gating

HardGate, Top5-TaskRR and TopK-TaskFilt, together with their LDA counterparts HardGate-LDA and SoftGate-LDA, rely on a single global Gaussian per task to either restrict the candidate set or to project features into a task-specific subspace. On CIFAR-100 (Table 4.4) all four strategies underperform the unrestricted Bayesian classifier when paired with the diagonal branch, by margins of 3–13 accuracy points. The cause is structural: when the diagonal Gaussian is already well calibrated, the additional volume contribution of a task-level prior introduces noise rather than information. The full-covariance branch is more forgiving (the volume term carries more information there) but still does not benefit from hard gating: TopK-TaskFilt attains 88.5 %, identical to the unrestricted Full-Cov baseline. The conclusion is that hard task gating is brittle on the in-distribution benchmark and should be reserved for settings in which the task identity can be recovered with high confidence—a condition that rarely holds in class-incremental learning.

5.3.2 Robustness of Additive Log-Domain Fusion

The two soft strategies SoftTaskFus and TaskMeanShr exploit the same task-level information as the hard variants but combine it additively in the log-domain. On every benchmark and every covariance branch they outperform the corresponding hard variants by 5–15 accuracy points, and on Caltech-256 they achieve the highest non-prompt accuracy reported in this thesis (95.0 % with Full-Cov plus SoftTaskFus). The robustness of additive fusion is structural: the task likelihood acts as a Bayesian prior that reinforces coherent decisions but cannot veto a strong class-level signal. This behaviour is a textbook example of well-behaved hierarchical evidence aggregation and provides the cleanest illustration in this work of the principle that hierarchical evidence should be combined multiplicatively (i.e. additively in the log-domain) rather than gated discretely.

5.3.3 Discriminative Refinement via Cosine and Min–Max Fusion

Top5-CosRR and OptFusion-MM combine the Mahalanobis log-likelihood with the angular similarity between the input feature and the class centroid. They are consistently competitive:

OptFusion-MM (the Optuna-tuned min–max fusion) attains the best A_{10} on three of the four benchmarks (CIFAR-100, Caltech-256 and Joint with Full-Cov). The mechanism is geometric: as shown in Section 5.1.2, DINOv2 produces features that are approximately isotropic at the global scale, so the cosine similarity captures directional information that the absolute log-likelihood discards (the unit-norm direction of the feature vector). The two scores are therefore complementary, and the Optuna search converges to fusion weights $\alpha \in [0.30, 0.85]$ depending on the benchmark, with the cosine term contributing most on the heavy-tailed benchmarks and the log-likelihood term dominating on CIFAR-100. The simpler "re-rank the Top-5 candidates by cosine" variant (Top5-CosRR) is a strong runner-up and exposes no hyperparameter, which makes it an attractive default whenever a hyperparameter search is impractical.

5.3.4 Temperature Scaling as Posterior Smoothing

The TempScale strategy (Section 3.6 of the methodology, denoted MahalanobisT) does not target accuracy but the conjoint goals of low forgetting and good calibration. Across the four benchmarks it produces the lowest forgetting in the Full-Cov branch on CIFAR-100 (4.05 %) and is consistently among the three best in F_{10}. The temperature parameter T acts on the log-likelihood by contracting the log-gap between the top-1 and top-2 candidates, with the dual effect of stabilising the maximum-a-posteriori decision as classes accumulate and of reducing the ECE, as confirmed by the additional metrics in Section 5.4.2. Like OptFusion-MM, TempScale benefits from per-dataset tuning via Optuna; the optimal T^* ranges from 0.02 on CIFAR-100 (where the unscaled log-gaps are already moderate) to 5.7 on ImageNet-R (where the log-gaps grow by several orders of magnitude).

5.3.5 The Joint Benchmark as a Diagnostic Lens

Table 4.7 is the most informative single result table in this thesis, because the long-horizon Joint stream amplifies the differences between strategies that would otherwise saturate on the 10-task benchmarks. The unrestricted Diag-Cov classifier reaches $A_{30} = 78.32\,\%$ and the unrestricted Full-Cov reaches $A_{30} = 78.65\,\%$. However, the Full-Cov branch *collapses* to 47–53 % under HardGate, Top5-TaskRR, TopK-TaskFilt, SoftTaskFus and TaskMeanShr, while the same strategies stay at 78–79 % in the Diag-Cov branch. The mechanism is the determinant ill-conditioning of the high-dimensional Full-Cov estimator (Section 5.1.3): when the task-level Gaussian inherits the same rank deficiency, hard gating amplifies its misrouting errors dramatically. The single Full-Cov re-ranker that survives is OptFusion-MM, which attains $A_{30} = 81.94\,\%$ and improves the unfused classifier by 3.3 accuracy points—the largest gain produced by any re-ranker on any benchmark in this thesis. TempScale recovers $A_{30} = 69.7\,\%$, an 18-point recovery over the worst hard-gating result, without any additional tuning beyond the Optuna search of T. These two strategies therefore play the dual role of accuracy enhancer (OptFusion-MM) and robustness mechanism (TempScale).

5.3.6 Summary Taxonomy of Re-ranking Strategies

Aggregating the evidence above, the ten re-ranking strategies fall into three clearly distinguishable groups:

- **Recommended strategies:** SoftTaskFus, TaskMeanShr, Top5-CosRR, OptFusion-MM and TempScale. These rarely degrade the unrestricted baseline and frequently improve it, with absolute gains of up to 3.3 points on the Joint stream.
- **Brittle strategies:** HardGate, Top5-TaskRR and TopK-TaskFilt. These restrict the candidate set on the basis of a single Gaussian decision and amplify task-mis-routing errors; they should be reserved for scenarios with a small number of well-separated tasks.
- **Marginal strategies:** HardGate-LDA and SoftGate-LDA. The added LDA subspace projection improves their robustness slightly relative to their non-LDA counterparts, but they do not beat the simpler additive-fusion variants and entail a higher computational cost.

5.4 Transversal Findings: Backbone, Calibration and Cost

This section discusses four transversal issues that emerge from the experiments but do not fit naturally in the per-method analyses of Sections 5.2 and 5.3: the dependence of the framework on the choice of backbone, the probabilistic calibration of its predictions, its computational footprint and a set of limitations that point to natural extensions.

5.4.1 Sensitivity of the Framework to the Choice of Backbone

Table 4.8 indicates that the choice of backbone is the single largest lever for the framework. DINOv2 dominates the supervised alternatives by 5–35 accuracy points on CIFAR-100, ImageNet-R and Joint; ViT-base/16 in particular collapses to 43–46 % on ImageNet-R. The empirical pattern correlates with the Gaussianity statistics of Table 4.1: DINOv2 produces approximately Gaussian, well-conditioned embeddings, whereas supervised ViTs trained on a fixed labelled corpus tend to produce more anisotropic features that violate the modelling assumption. Caltech-256 is the only exception: the three backbones cluster within four accuracy points, with ViT-medium narrowly ahead. This pattern is consistent with the interpretation that Caltech-256 is sufficiently easy on the visual side that the bottleneck moves from the encoder to the downstream classifier; the choice of backbone matters less when the categories are already linearly separable in the feature space.

5.4.2 Probabilistic Calibration as a Quality of the Volume Term

The ECE results of Table 4.9 provide an independent view of the volume–shape trade-off discussed in Section 5.2.1. The diagonal estimator is systematically better calibrated than the full-covariance one. The gap is marginal on CIFAR-100 (0.2 percentage points) but reaches 25.7 points on the Joint stream. The explanation is, once again, that the full-rank log-determinant in $d = 768$ produces extreme volume scores; once they are normalised through the softmax, the posterior collapses onto a handful of classes regardless of the actual quality of the prediction. Temperature scaling alone is insufficient to repair this collapse in the long-horizon regime, which is the reason why the methodology recommends Hybrid-Cov as the calibration-aware default rather than Full-Cov.

The complementary metrics reported in Table 4.10 make the diagnostic richer. The Diag-Cov branch behaves in a numerically reasonable regime on every benchmark: Total Confidence remains between 0.84 and 0.97, Confidence on Errors stays well below one (between 0.47 and 0.81), the Log-Gap between the top-1 and top-2 log-likelihoods is bounded by approximately 5 and 155, and the predictive entropy is non-trivial (0.07–0.61 nats). The Full-Cov branch, by contrast, saturates to a degenerate confidence regime on every benchmark: Total Confidence equals 1.0000, Confidence on Errors collapses to approximately one (so the model is just as confident when it is wrong as when it is right), entropy drops below 10^{-3} nats, and the Log-Gap explodes to values between 5×10^3 on ImageNet-R and 2.7×10^6 on the Joint stream, reaching 8.9×10^7 even on CIFAR-100. The numerical signature of this collapse is the same one identified in Section 5.1.3: the rank-deficient determinant produces astronomically peaked posteriors that are essentially deterministic regardless of input.

A particularly informative consequence of this analysis is that the Full-Cov branch attains *higher* final-task accuracy than Diag-Cov on CIFAR-100 (88.3 % versus 86.4 %) while being miscalibrated by eight additional ECE points. This is the cleanest empirical illustration in the thesis of the principle that classification accuracy and probabilistic calibration are largely orthogonal in the rank-deficient regime: the volume-driven peaking that destroys calibration does not necessarily destroy accuracy, provided the per-class shapes are still discriminative. On the harder benchmarks the rank-deficient determinant becomes actively harmful and Full-Cov loses accuracy on top of being miscalibrated, which is why Hybrid-Cov is the recommended default.

5.4.3 Computational Footprint and Practical Scalability

A side observation that is easy to overlook in the tables of Chapter 4 is that every method proposed in this thesis is closed-form. The training cost is dominated by the single forward pass through the frozen DINOv2 backbone and by the closed-form estimation of the per-class means and covariances. By contrast, the prompt-based competitors require gradient updates of the prompt pool at every incremental task. On CIFAR-100, adding a new task to our framework requires approximately two minutes on a single consumer-grade GPU, almost entirely spent on feature extraction; prompt-based methods reported in the literature typically require 10–30 minutes per task on comparable hardware. The proposed framework therefore offers a 5–15× speedup at training time, without any rehearsal storage requirement.

5.4.4 Limitations and Open Questions

The present evaluation has four limitations that deserve explicit acknowledgement. First, the Joint benchmark assumes that task boundaries are known at training time; relaxing this assumption (task-free continual learning) is a natural follow-up that would test the robustness of the additive task-level priors of Section 5.3.2. Second, the re-ranking strategies do not currently exploit the per-task Gaussianity statistics reported in Table 4.1; an adaptive fusion in which the weight α depends on the empirical kurtosis of the active classes is a promising extension. Third, the evaluation is restricted to visual classification: although the framework is in principle agnostic to the modality, an empirical assessment on text or speech embeddings would provide a valuable external test of the underlying Gaussian assumption. Fourth, a more speculative direction would be to extend the framework to open-world settings by treating the predictive entropy and Log-Gap of Table 4.10 as out-of-distribution scores; the results of Chapter 4 suggest that the Diag-Cov branch is the only one in which these quantities are meaningful.

5.5 Synthesis and Practical Recommendations

The experimental evidence collected in Chapter 4 supports four high-level conclusions that, taken together, justify the design choices of the proposed framework.

Finding 1: Gaussian class-conditional modelling is a viable design choice on modern self-supervised embeddings. The DINOv2 features satisfy the marginal-normality assumption on 96 % of the coordinates in CIFAR-100, 86 % in ImageNet-R and 75 % in Caltech-256, with the residual non-Gaussianity concentrated in the heavy-tailed Caltech-256 case. The single-coordinate Mahalanobis histograms (Figure 4.5b) confirm that the per-component distributions track the theoretical χ^2_1 density. The pairwise KL divergences of Figure 4.6 additionally show that the average distinguishability between class Gaussians is large and stable across all class counts, while the minimum KL diagnoses the small set of intrinsically confusable pairs that account for the residual forgetting. A classifier built directly on top of these statistics achieves performance competitive with prompt-based state of the art while remaining closed-form, deterministic and gradient-free.

Finding 2: The Hybrid covariance is the most useful single estimator developed in this thesis. By combining the discriminative power of the full-rank Mahalanobis distance with the calibration properties of the diagonal log-determinant, the Hybrid estimator outperforms every alternative on the heavy-tailed and long-horizon benchmarks (Caltech-256 and Joint), and matches the best linear shrinkage estimators on CIFAR-100. The linear shrinkage estimators (OAS, Ledoit–Wolf) repair the covariance for estimation purposes but flatten the eigenvalue spectrum that the classifier exploits, and they are dominated by Hybrid in every regime that is not trivially in-distribution.

Finding 3: Additive log-domain fusion is the only re-ranking pattern that improves the baseline reliably. The strategies SoftTaskFus, TaskMeanShr, OptFusion-MM, Top5-CosRR and TempScale consistently improve, or at worst match, the unfused baseline, with the largest absolute gain (3.3 accuracy points on Split Joint with Full-Cov) achieved by OptFusion-MM. Hard-gating strategies (HardGate, Top5-TaskRR, TopK-TaskFilt), by contrast, destroy accuracy whenever the task-level Gaussian is mis-calibrated—a condition that arises precisely on the benchmarks where re-ranking would otherwise be most needed.

Finding 4: The framework's accuracy ceiling is set by the upstream representation, not by the downstream classifier. The empirical gap between DINOv2 and supervised ViT variants in Table 4.8 is larger than any gap induced by the choice of covariance estimator or re-ranker. The pragmatic implication is that future improvements should target the upstream representation rather than the downstream classifier: a more discriminative, better-conditioned encoder will lift the entire framework without requiring methodological changes.

Practical recipe. The four findings outline a concrete recommendation for building incremental classifiers on top of frozen foundation models: estimate a Hybrid-Covariance Gaussian per class, optionally re-rank candidates with the Optuna-tuned min–max cosine fusion (OptFusion-MM), and—whenever calibrated probabilities are required—wrap the score in a temperature-scaled softmax (TempScale). This recipe is simple, scalable, gradient-free and empirically competitive with the prompt-based state of the art. It provides a clean foundation for the open-set and detection extensions discussed in Chapter 6.

Chapter 6

Conclusion

6.1 Future Work

At this stage, the integration of Open World Discovery appears to be a straightforward extension of the existing framework. Since our approach provides a certainty score, implementing an Out-of-Distribution (OOD) threshold allows the system to flag unknown classes with minimal architectural changes. This addresses the objective of moving beyond a closed-set classifier toward an adaptive, "discovery-ready" system.

In contrast, the transition to Real-Time Object Detection requires a more strategic approach. While previous experiments utilising a Region Proposal Network (RPN) [49] yielded high localisation accuracy, the computational overhead of a dual-model pipeline (RPN + Feature Extractor) compromises the inference speed. By leveraging the spatial tokens already present in the DINOv2 backbone, we can identify regions of interest using simple mathematical decomposition. This ensures that the system remains lightweight and consistent with the unsupervised nature of the feature extractor.

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